

Intensity Transformation and Spatial Filtering

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Image enhancement

Image enhancement is the procedure of **improving the quality and information content** of original data before processing. Common practices include contrast enhancement, spatial filtering etc.

Domain of image processing

- ❑ Spatial Domain (Direct operation on image)
- ❑ Frequency domain (Operation after Fourier transformation)

Spatial processing

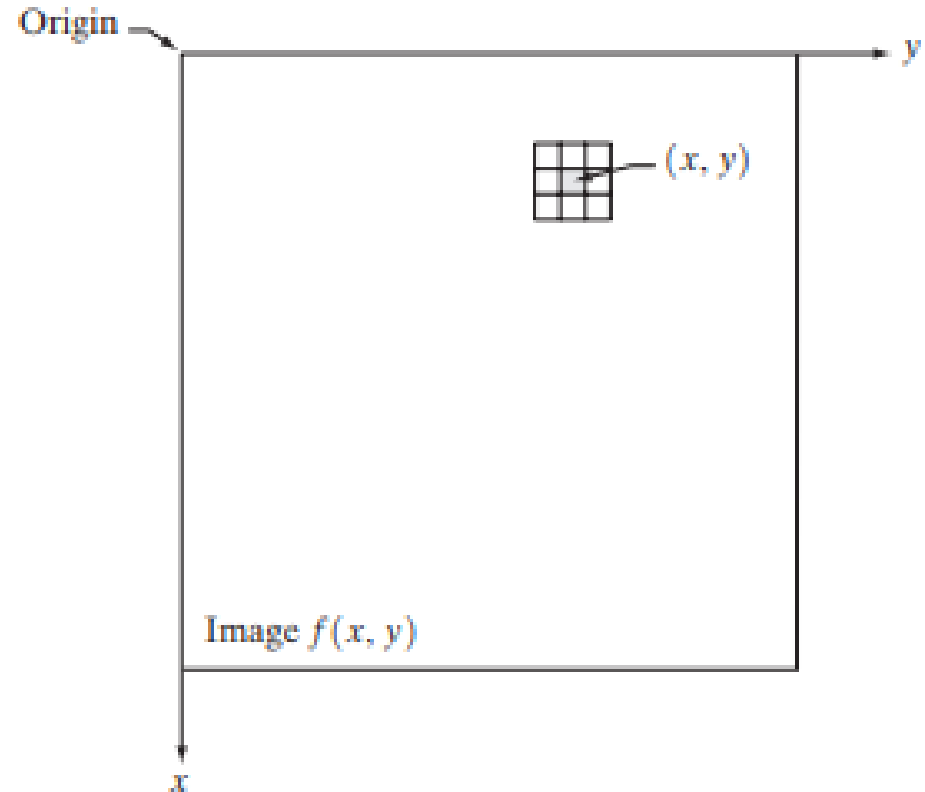
$$g(x,y) = T [f(x,y)]$$

- $f(x,y)$? Input Image
- $g(x,y)$? Output image
- T ? Operation/ Transform method

Spatial processing

$$g(x,y) = T [f(x,y)]$$

- ❑ $f(x,y)$? Input Image
- ❑ $g(x,y)$? Output image
- ❑ T ? Operation/ Transform
method defined over the
neighbourhood of point(x,y)



Spatial processing

For smallest neighbourhood of size $1 * 1$, g depends only on value of f at a single point (x,y) and T becomes a intensity transformer function. Also called gray level mapping.

Hence for simplicity of notation it is described by

$$s = T(r)$$

Here s and r denotes intensity of g and f at a point (x,y)

Spatial processing

- Intensity transformation

 - (Operates on single pixel)

- Spatial Filtering

 - (Operates in neighbourhood)

Intensity Transformation

Intensity transformations are applied on images for contrast manipulation or image thresholding. These are in the spatial domain, i.e. they are performed directly on the pixels.

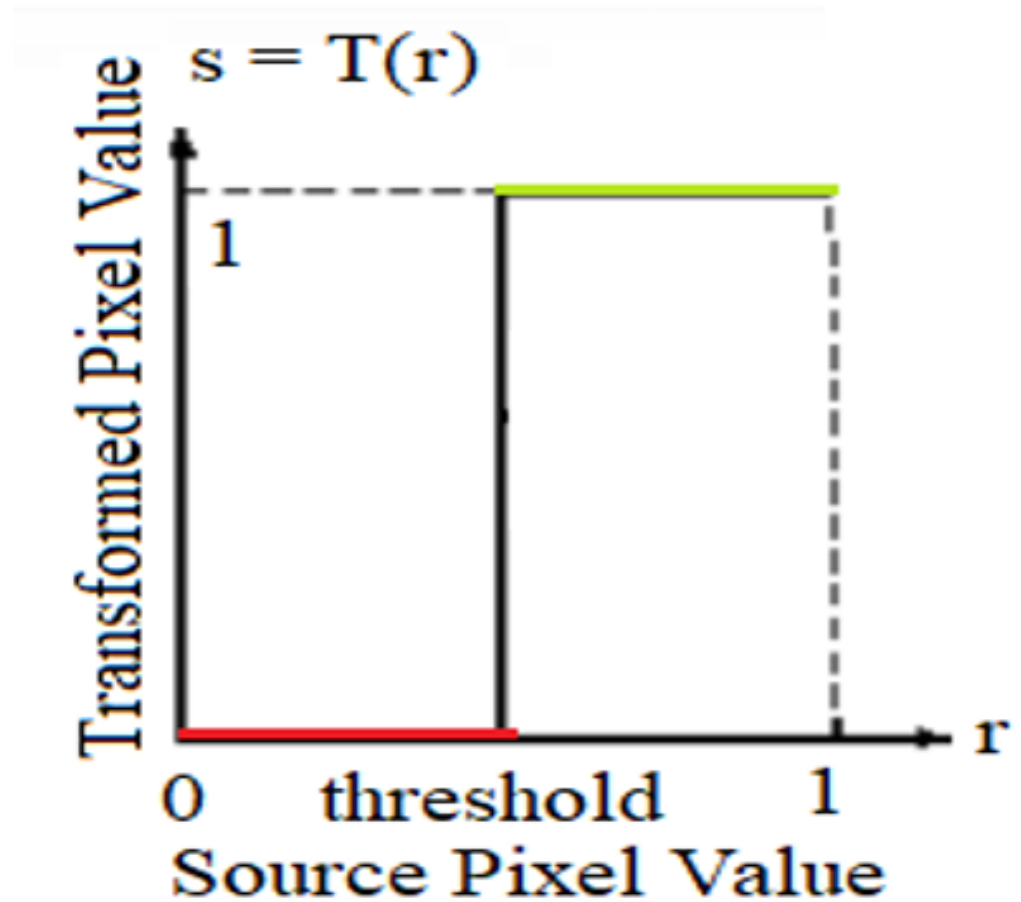
Simplest off all image processing techniques.

Intensity Transformation

Thresholding:

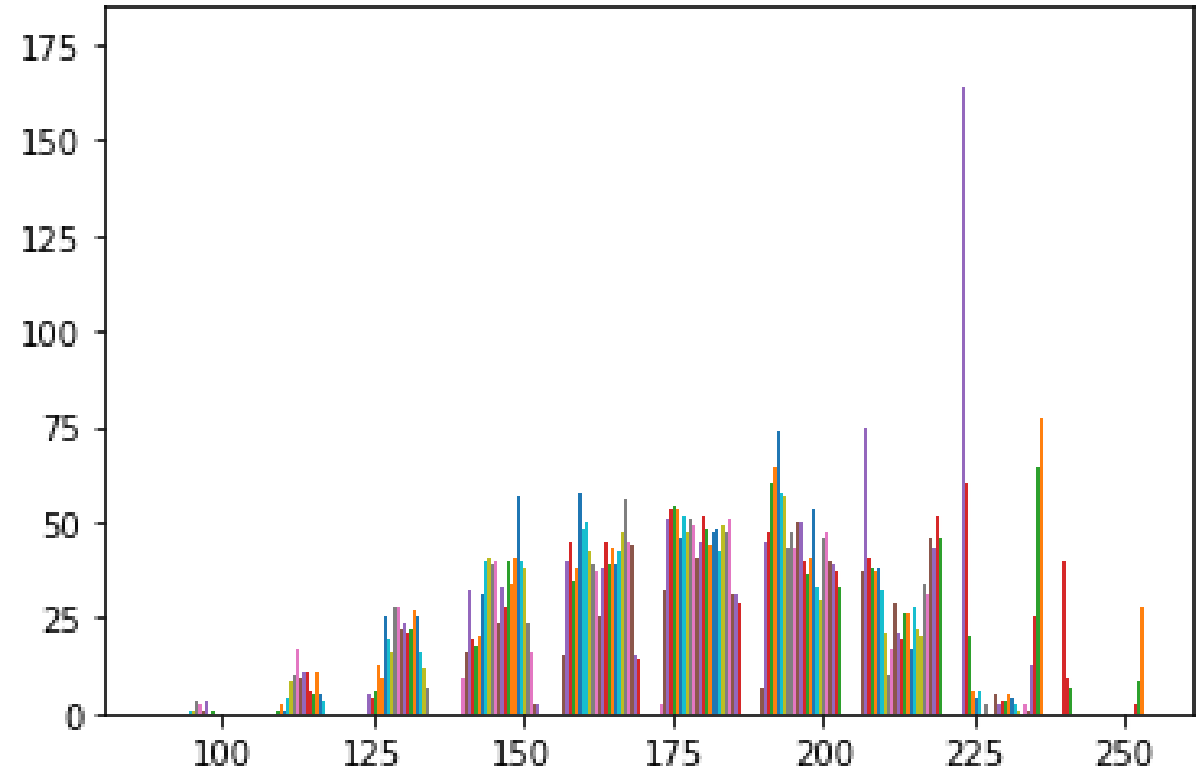
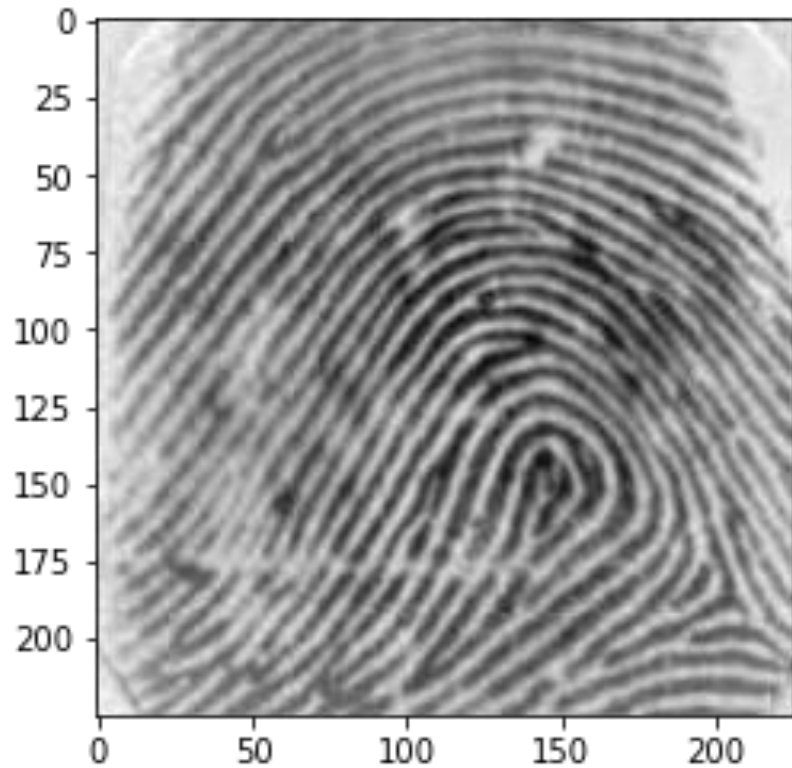
$s = 0$ if $r > \text{Threshold}$

$s = 255$ if $r \leq \text{Threshold}$



Intensity Transformation

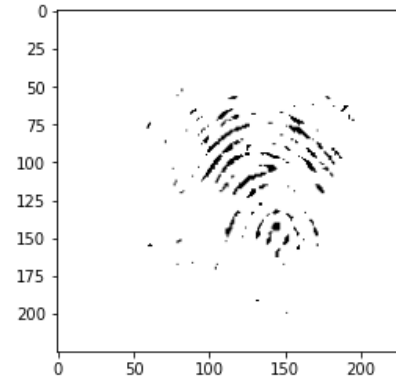
Thresholding



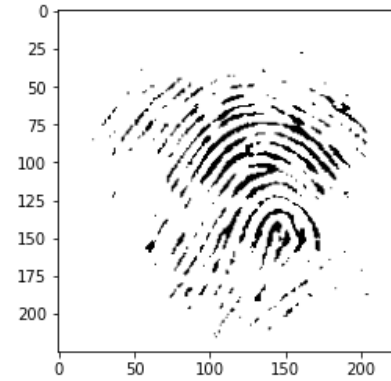
Intensity Transformation

Thresholding

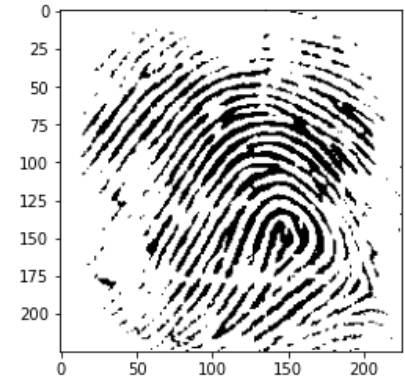
$T = 120$



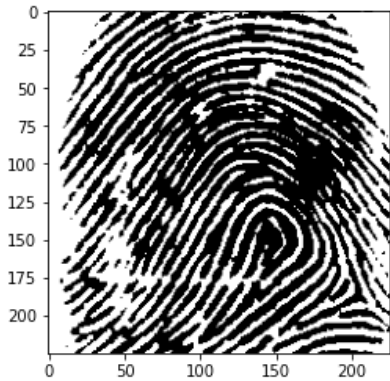
$T = 135$



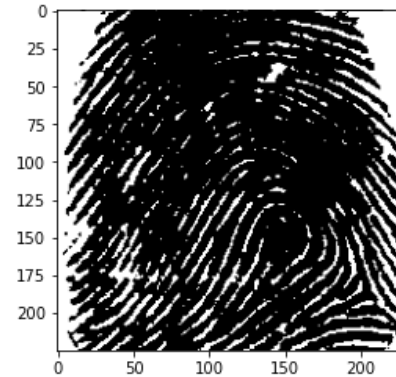
$T = 155$



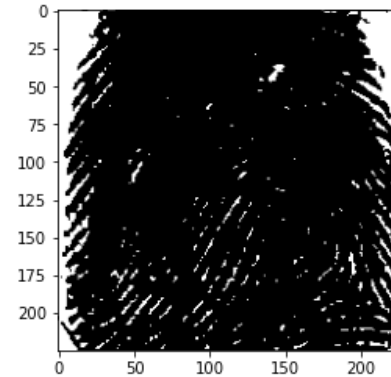
$T = 187$



$T = 205$

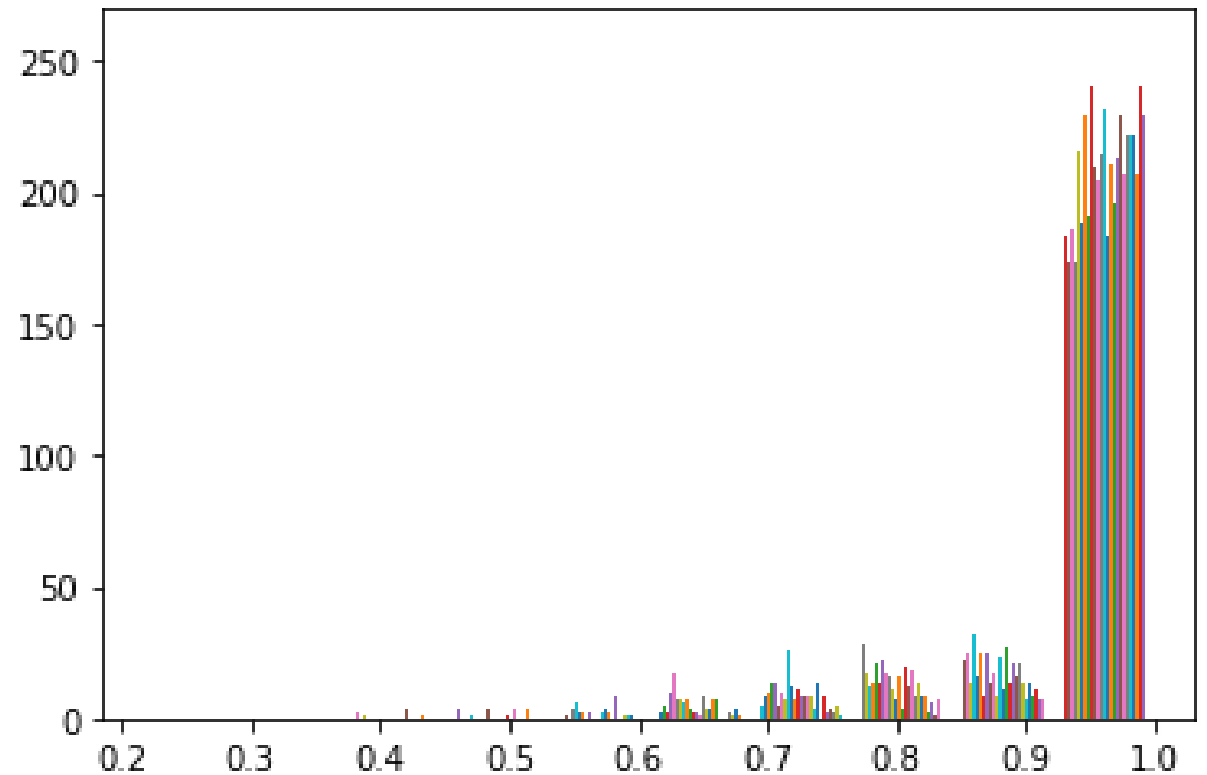
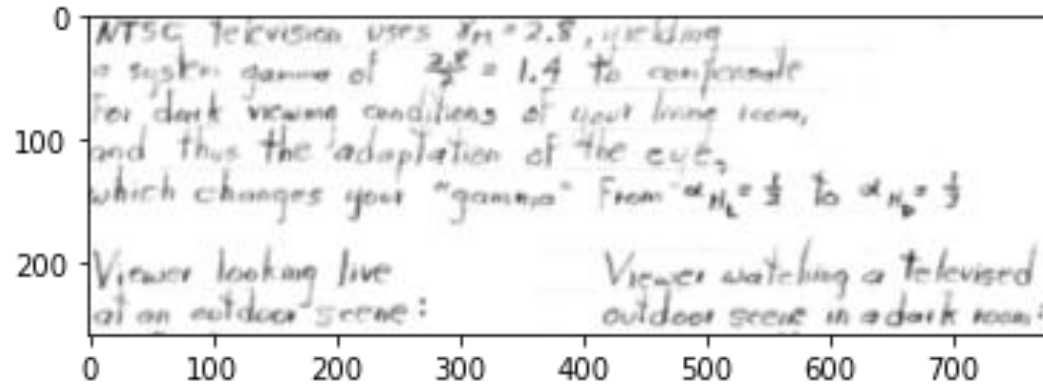


$T = 220$



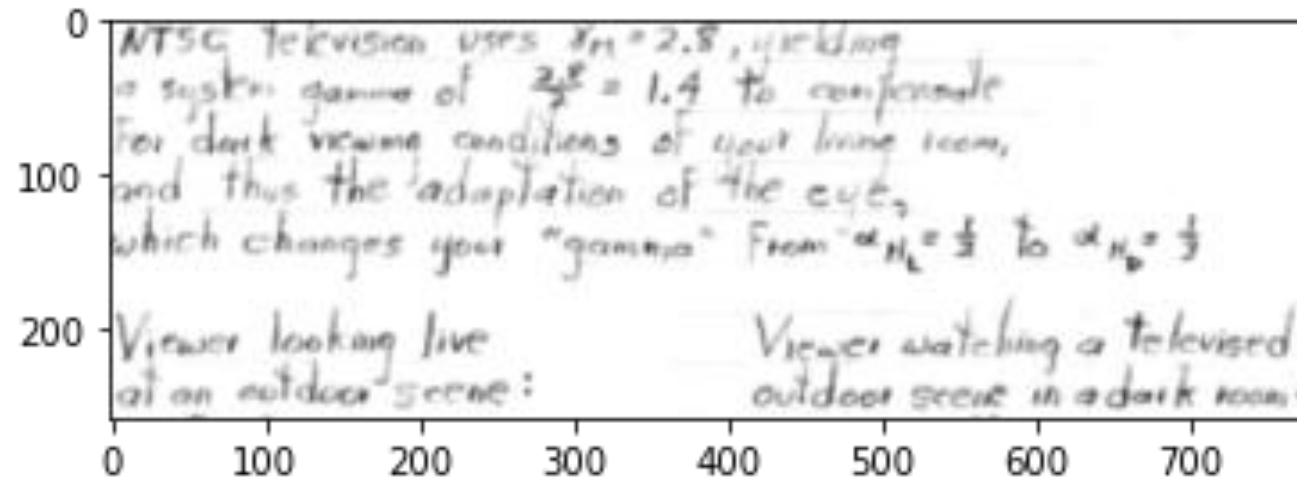
Intensity Transformation

Thresholding

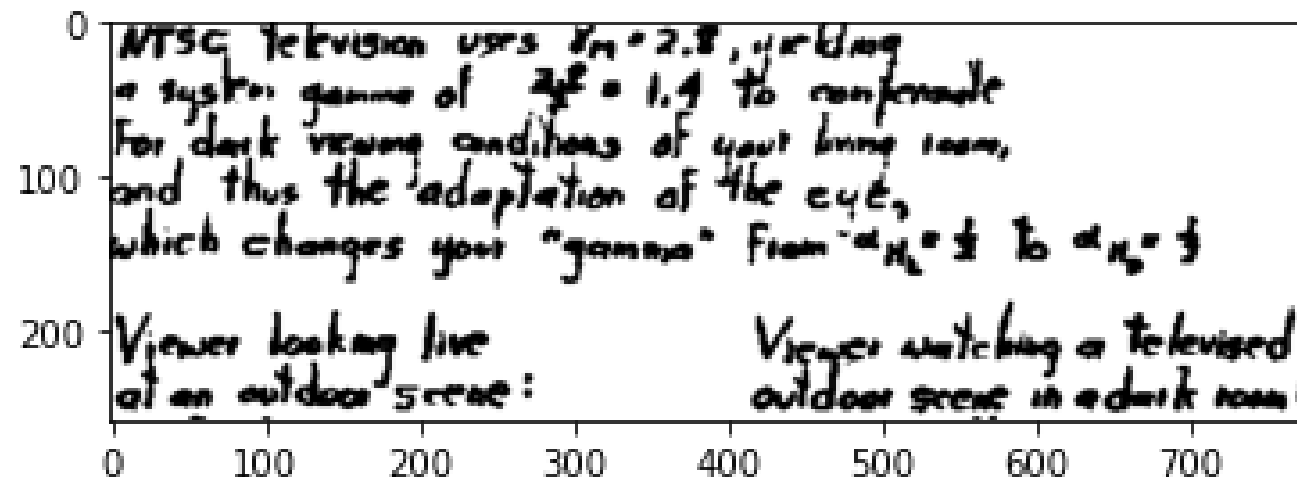


Intensity Transformation

Thresholding



T = 0.92



Intensity Transformation

Some basic subjective quality measurement terminologies:

1. Brightness
2. Contrast
3. Sharpness

Intensity Transformation

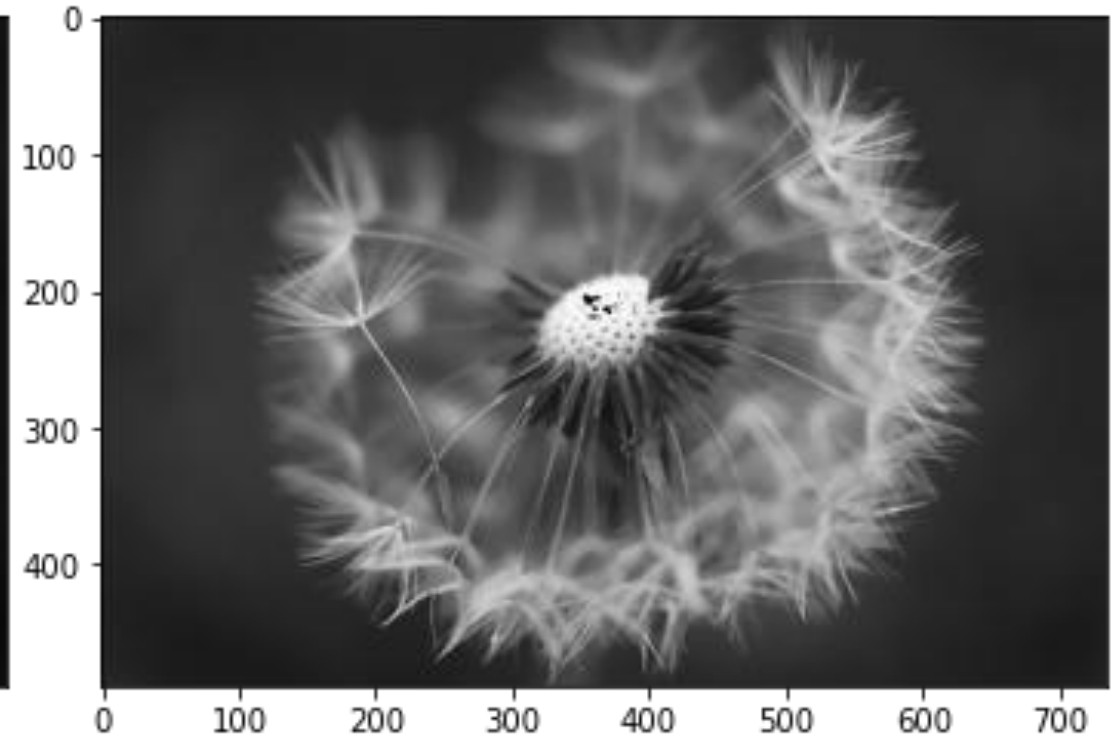
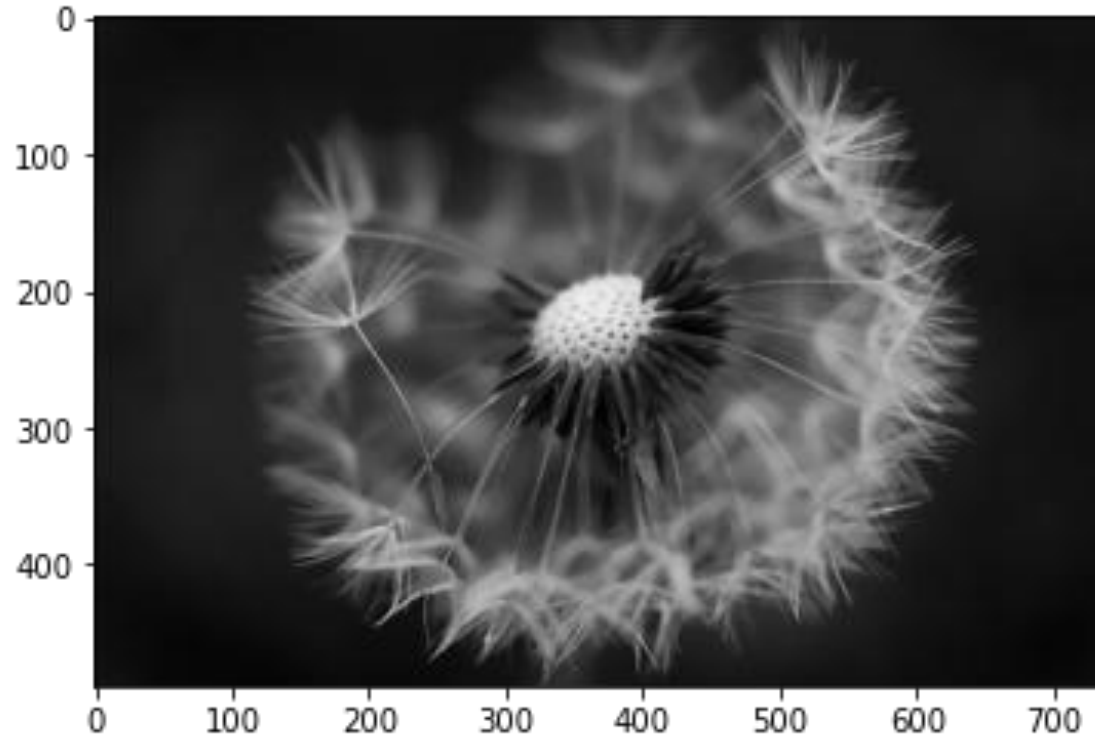
Brightness: In terms of digital image processing, brightness is more properly described as the measured intensity of all the pixels comprising an ensemble that constitutes the digital image.

Traditional brightness simply brightens the entire image from the shadows to the highlights equally.

Intensity Transformation

Brightness

$\text{img} + 20$

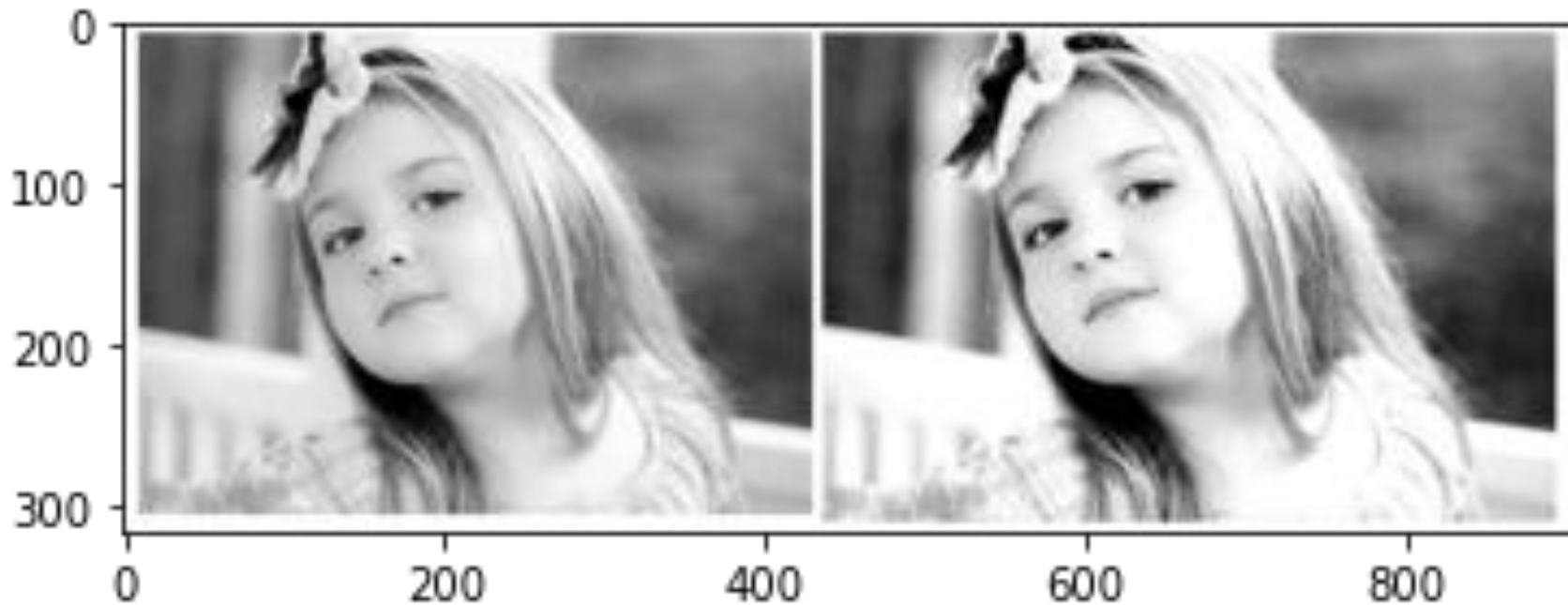


Intensity Transformation

Contrast: Contrast makes an object **distinguishable**. It is the **separation between** the **darkest** and **brightest** areas of the image. **Increase contrast** and you increase the separation between dark and bright, making shadows darker and highlights brighter.

Intensity Transformation

Contrast



Low contrast

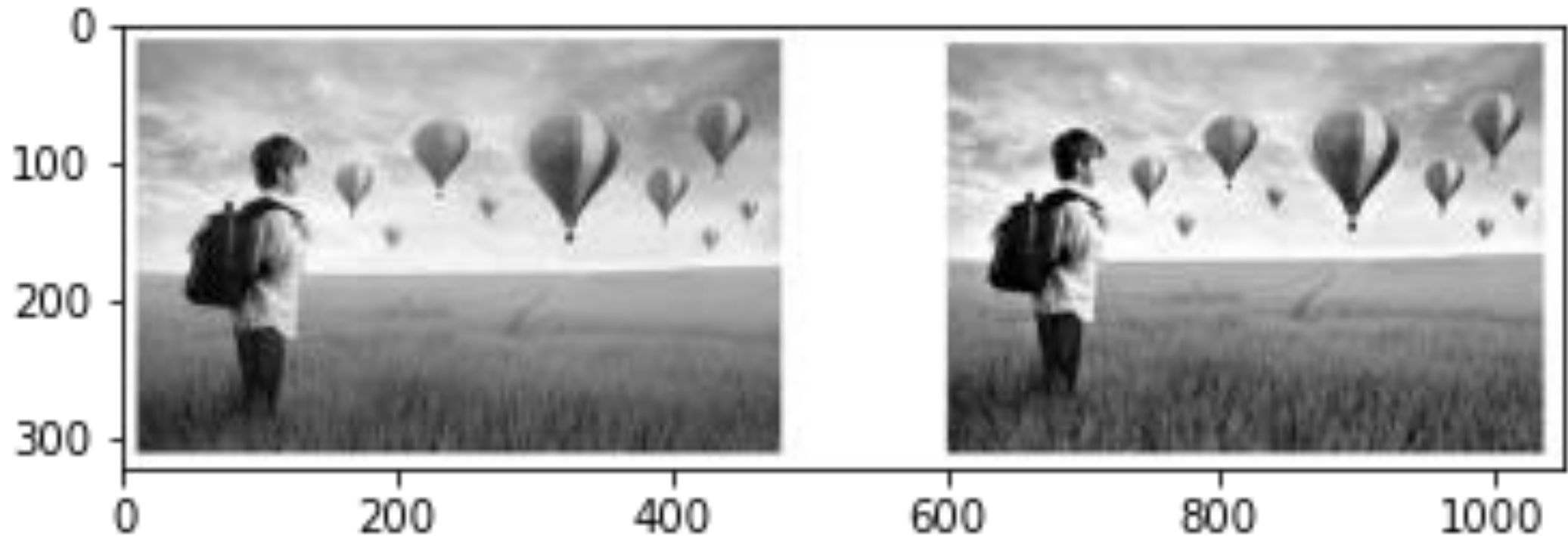
High contrast

Intensity Transformation

Sharpness: Sharpness can be defined as **edge contrast**, that is, the contrast along edges in a photo. When we increase sharpness, we increase the contrast only along/near edges in the photo while leaving smooth areas of the image alone.

Intensity Transformation

Sharpness



Some basic Intensity transformation Functions

There are three types of transformation:

- 1.Linear

- 2.Logarithmic

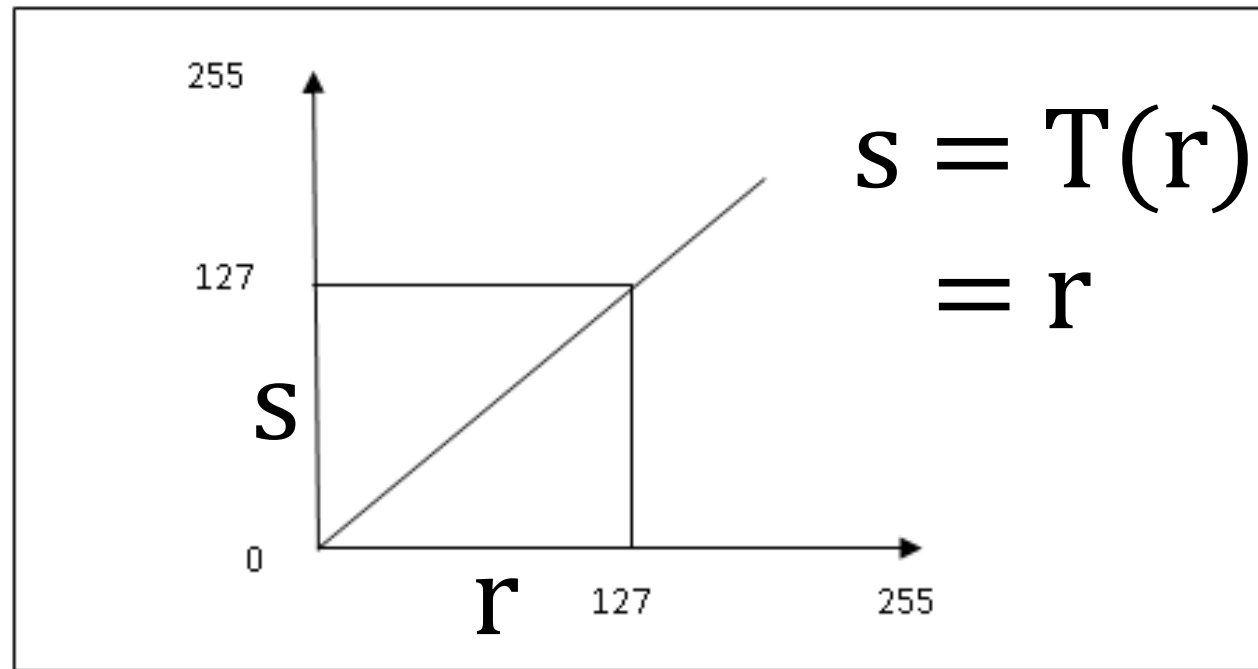
- 3.Power - law

Some basic Intensity transformation Functions

Linear

Identity transformation:

Each value of the image is directly mapped to each other values of the output image.



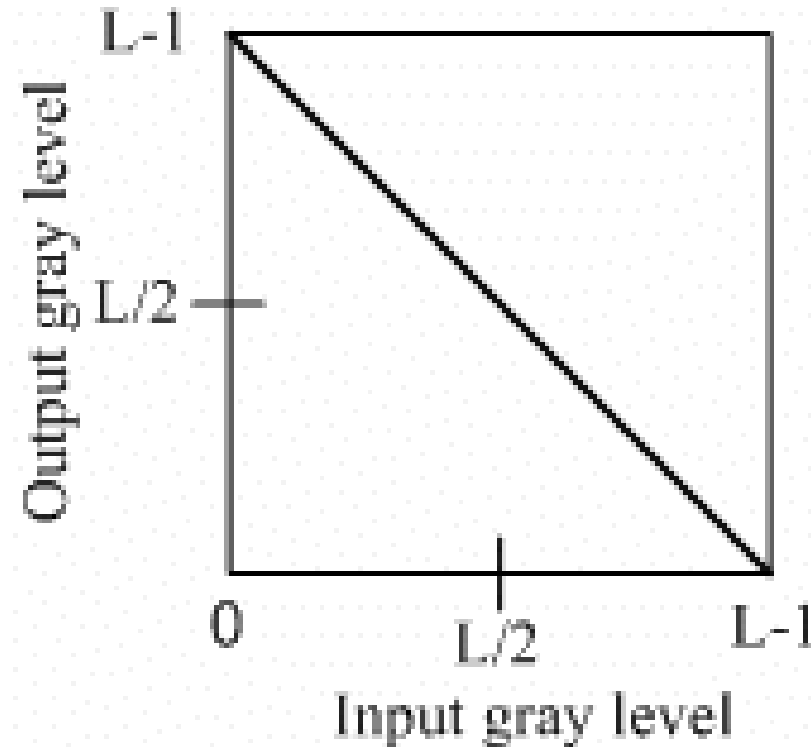
Identity transformation

Some basic Intensity transformation Functions

Linear

Negative transformation:

It is the opposite of identity transformation. Here, each value of the input image is subtracted from $L-1$ and then it is mapped onto the output image.

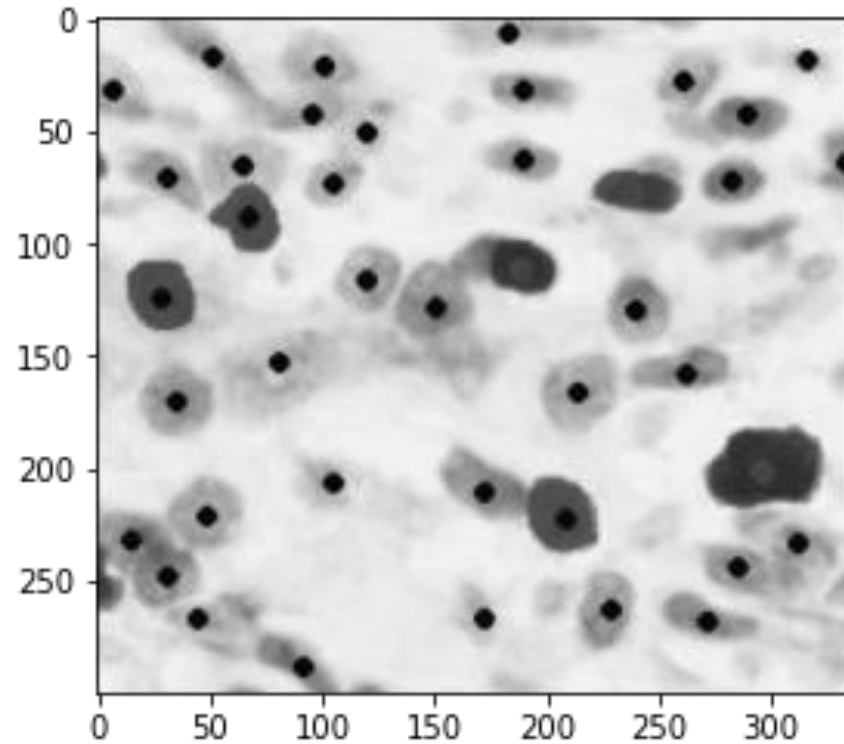
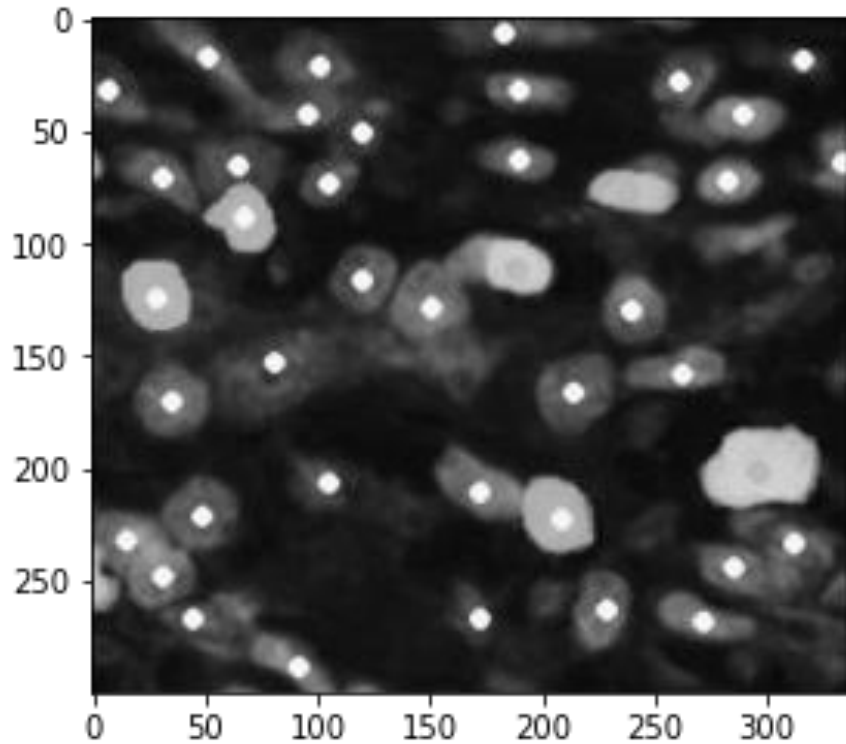


$$s = T(r) \\ = 255 - r$$

Some basic Intensity transformation Functions

Linear

Negative transformation



$$S = (L-1) - r$$

$$S = (8-1) - r$$

$$= 7 - r$$

Assuming the image to be 3-bit, we have $2^3 = 8$ levels.

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

$$r=0, S = 7-0 = 7$$

$$r=1, S = 7-1 = 6$$

$$r=2, S = 7-2 = 5$$

$$2^0 = 1$$

$$2^1 = 2$$

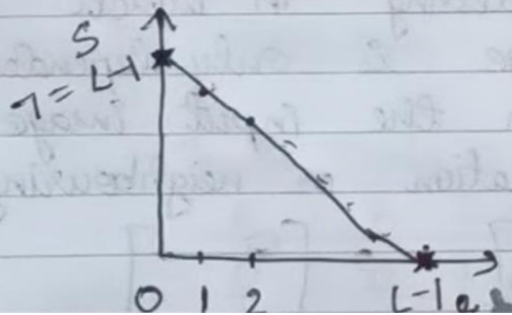
$$2^2 = 4$$

$$2^3 = 8 \quad 0-8$$

$$r=7, S = 7-7 = 0$$

$$n=3$$

$$L = 2^n = 2^3 = \underline{\underline{8}}$$



3	4	2	5
4	1	3	1
5	5	1	2
0	1	3	6

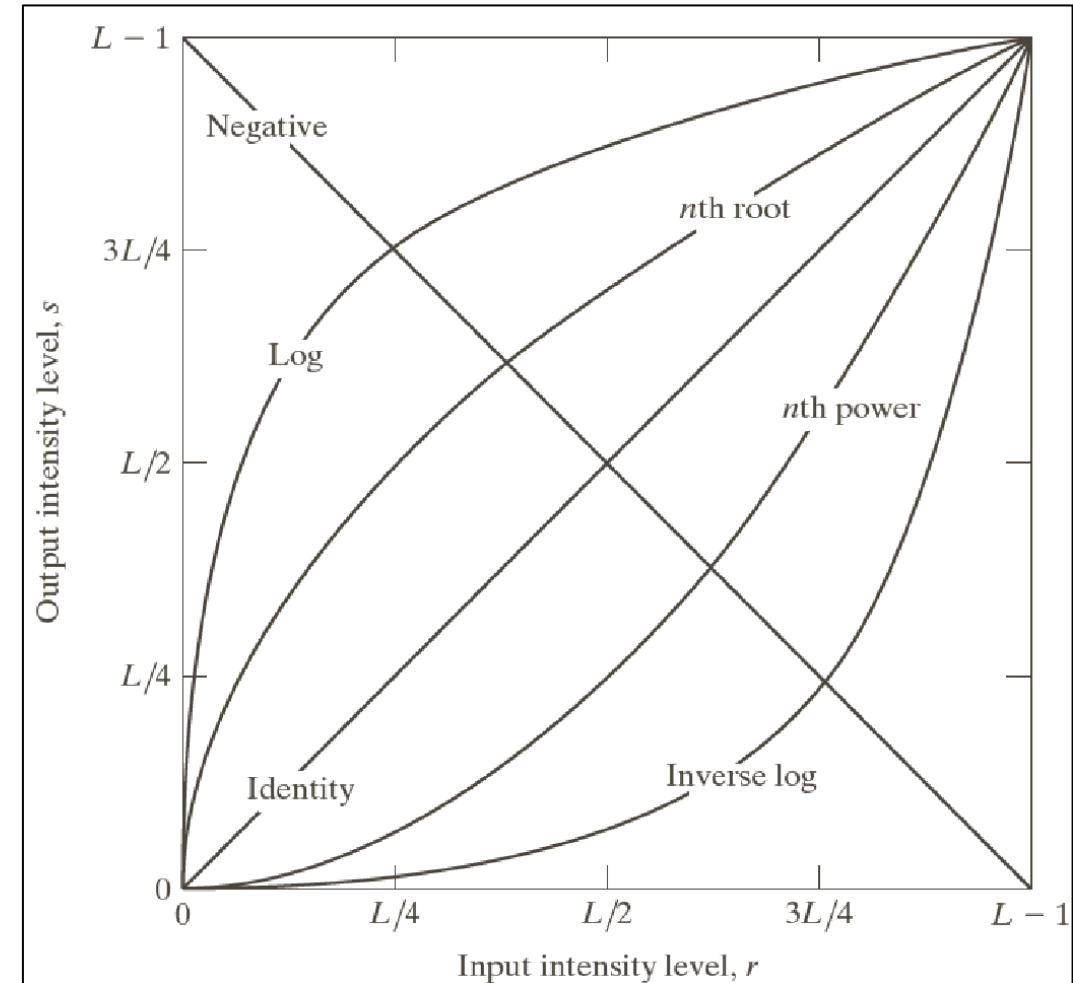
Digital Negative

Some basic Intensity transformation Functions

Logarithmic

Logarithmic transformation is divided into two types:

- ❑ Log transformation
- ❑ Inverse log transformation

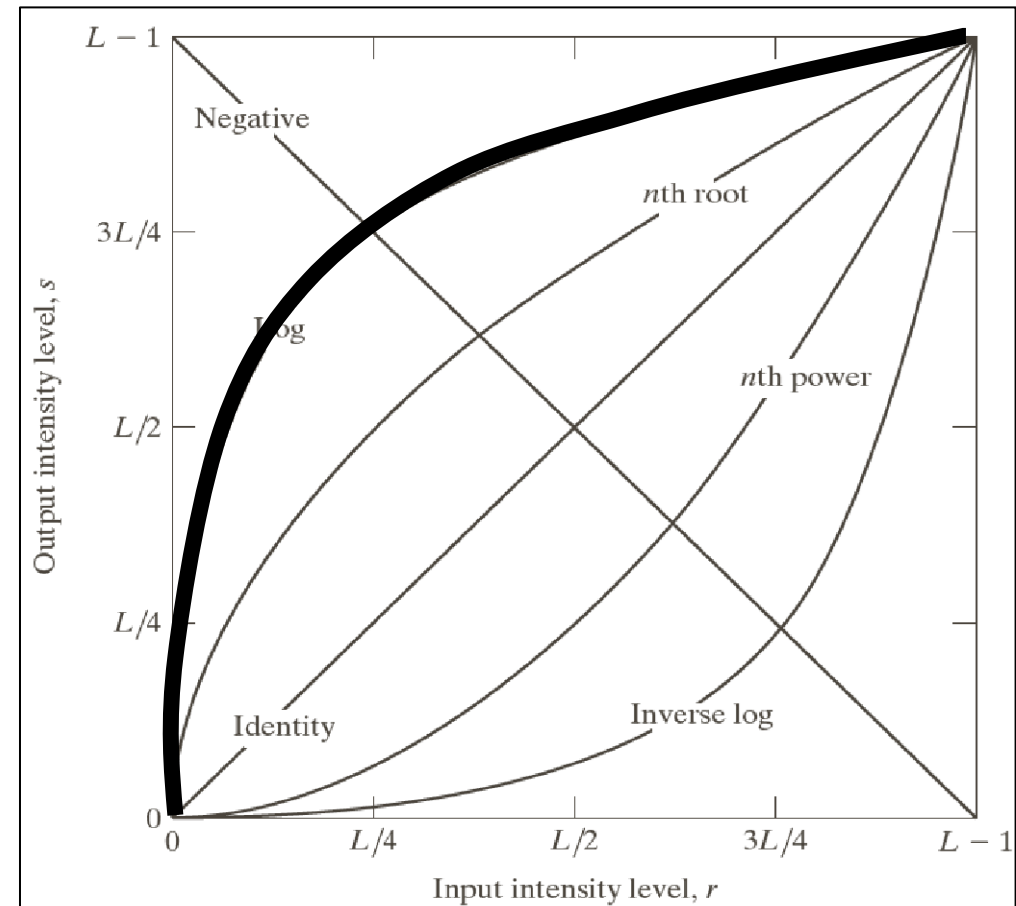


Some basic Intensity transformation Functions

Logarithmic

□ Log transformation

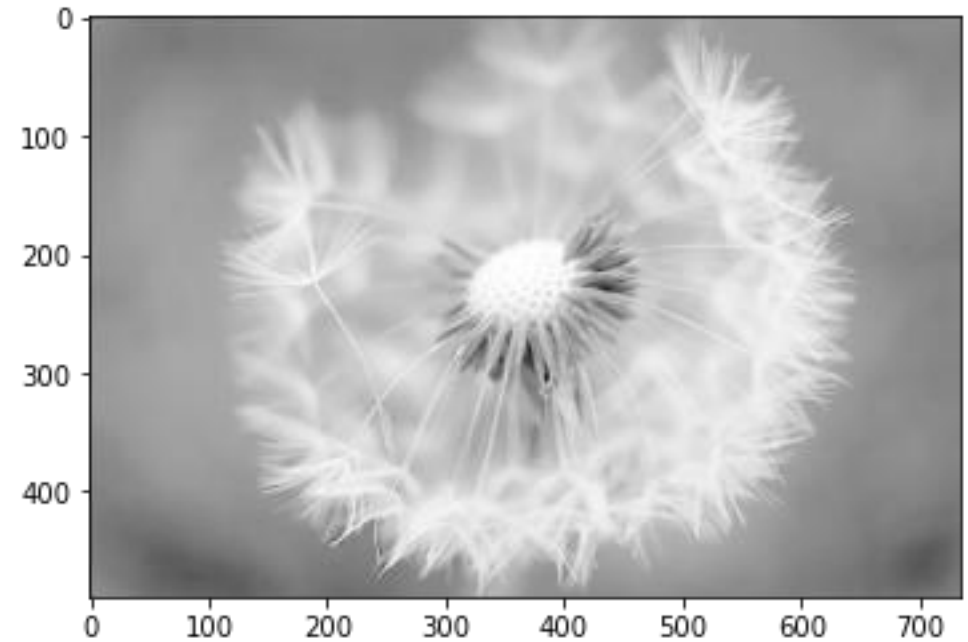
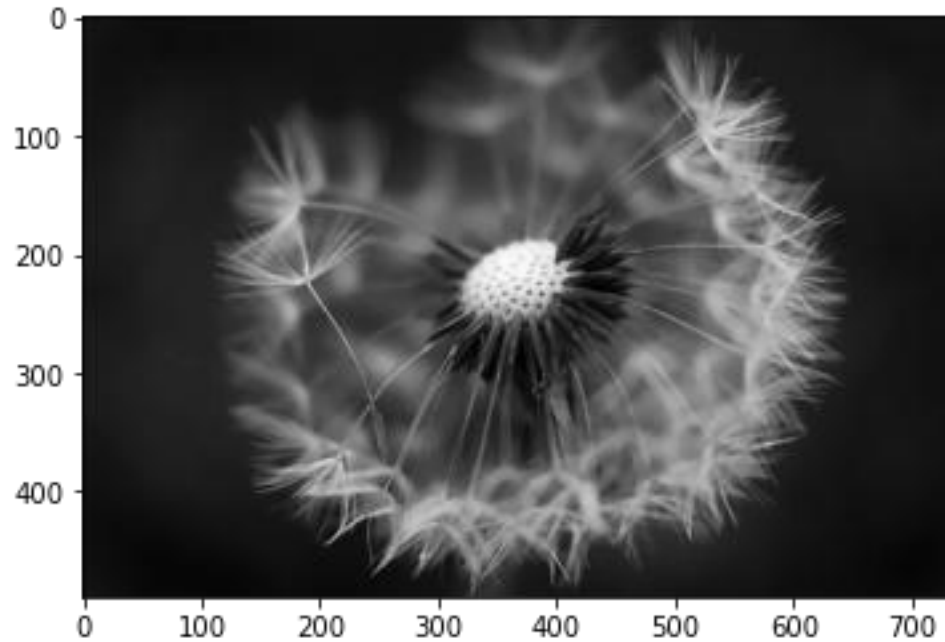
$$\begin{aligned}s &= T(r) \\ &= c * \log(1+r)\end{aligned}$$



Some basic Intensity transformation Functions

Logarithmic

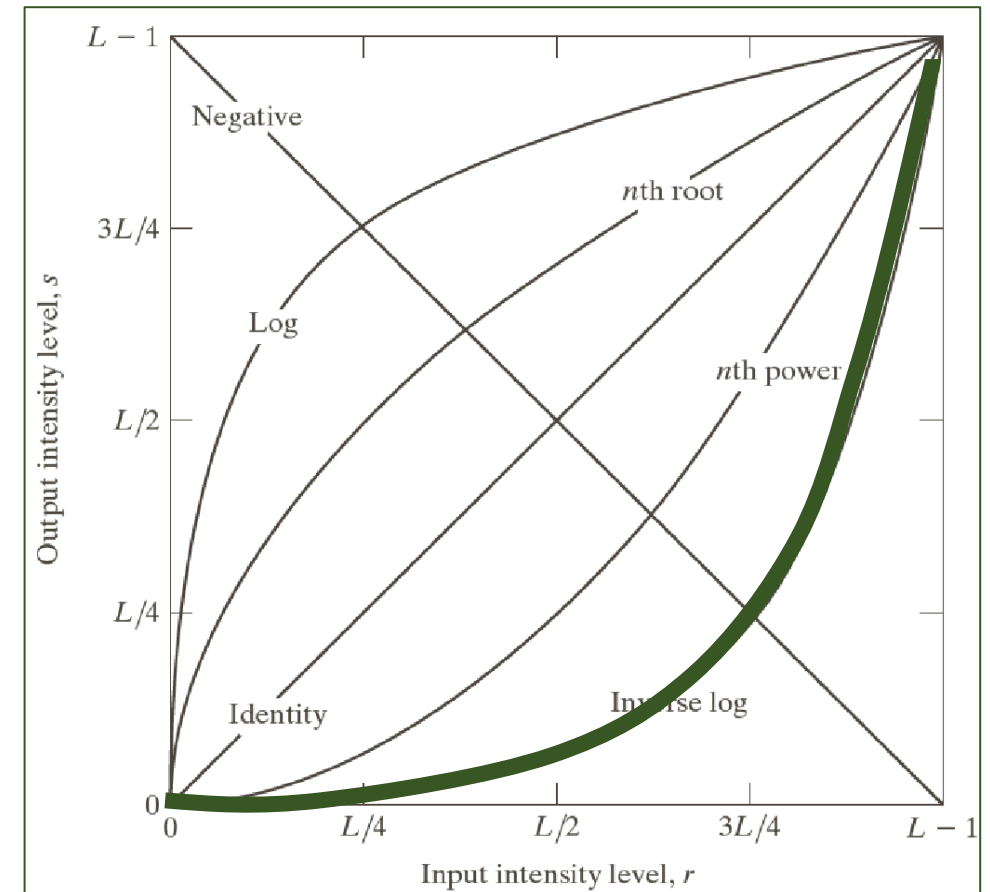
- ❑ Log transformation



Some basic Intensity transformation Functions

Logarithmic

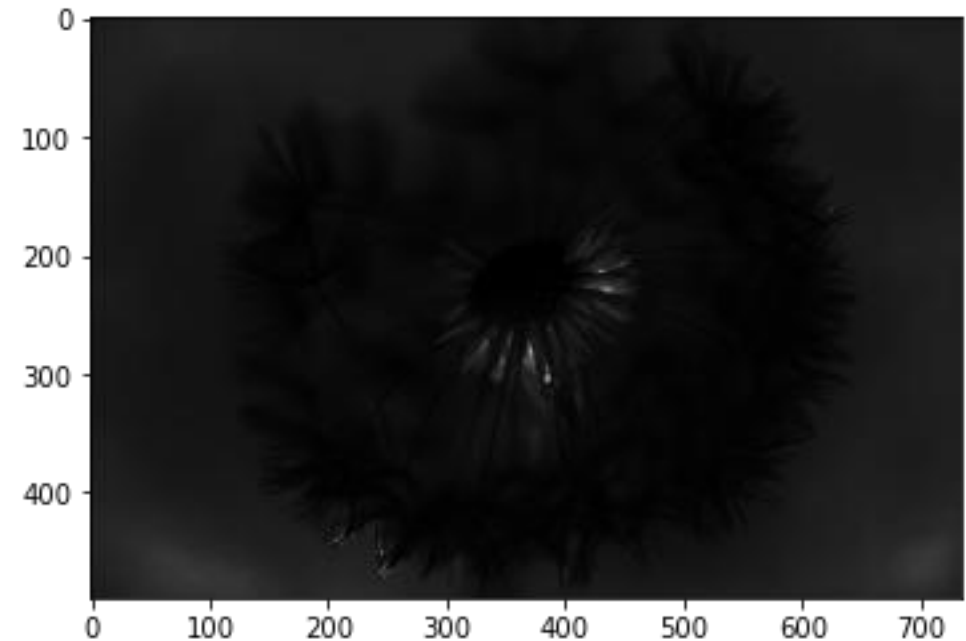
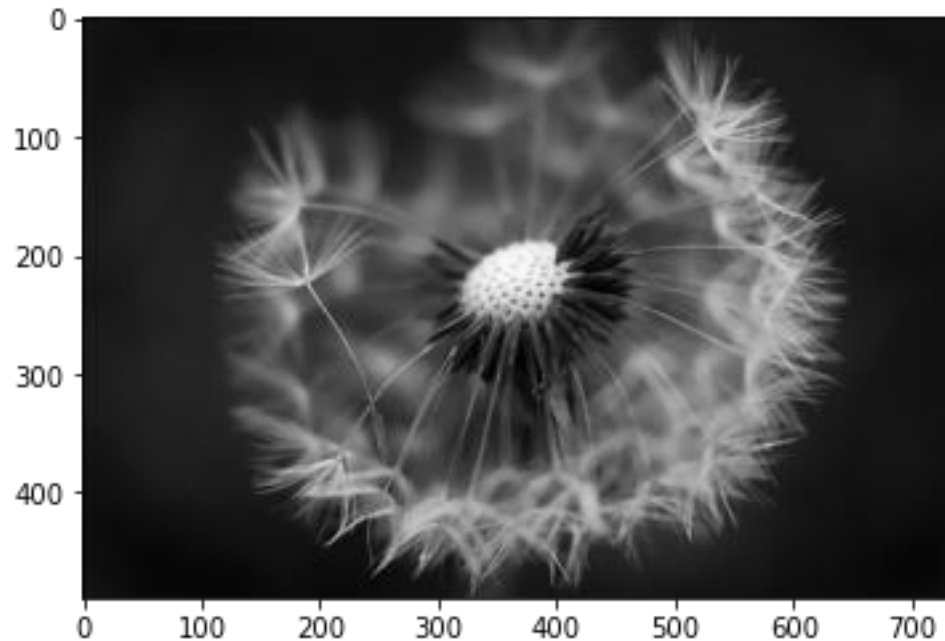
- ❑ Inverse log transformation



Some basic Intensity transformation Functions

Logarithmic

- ❑ Inverse Log transformation



Some basic Intensity transformation Functions

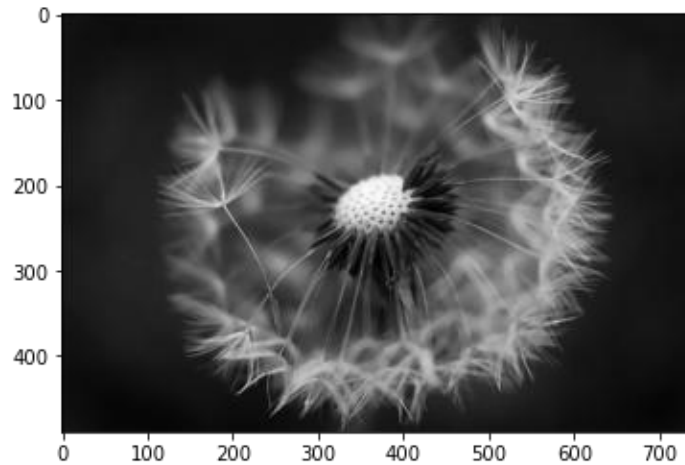
Power law (Gamma transformation)

$$\begin{aligned}s &= T(r) \\ &= c * r^\gamma\end{aligned}$$

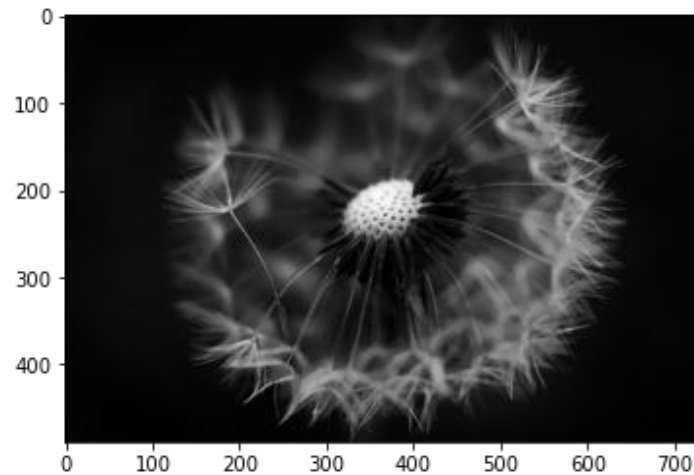
Some basic Intensity transformation Functions

Power law (Gamma transformation)

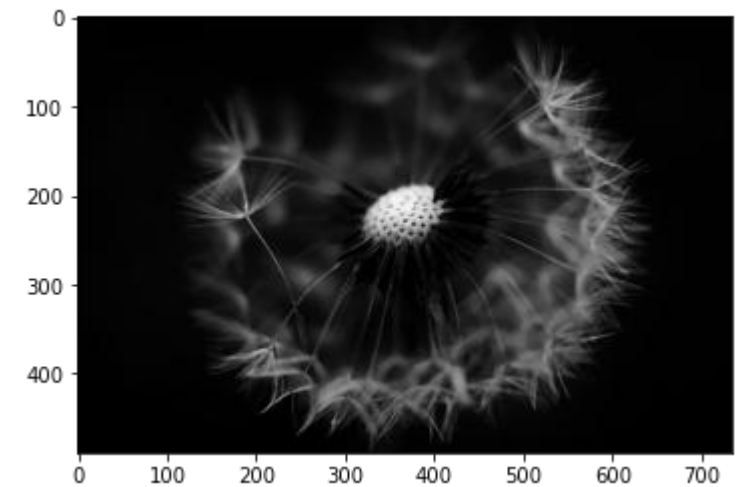
$\gamma = 1$



$\gamma = 1.5$



$\gamma = 2$



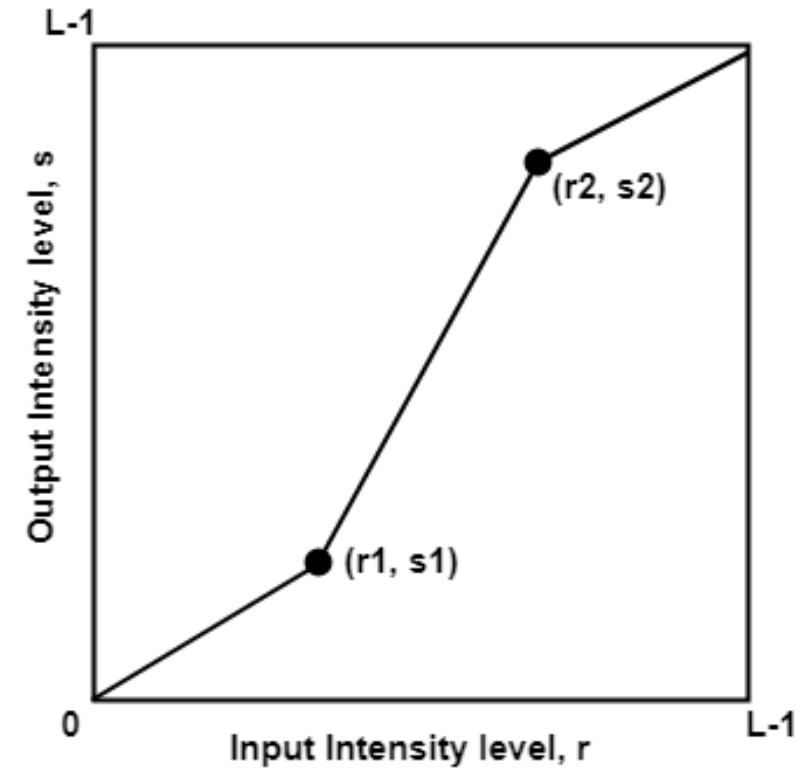
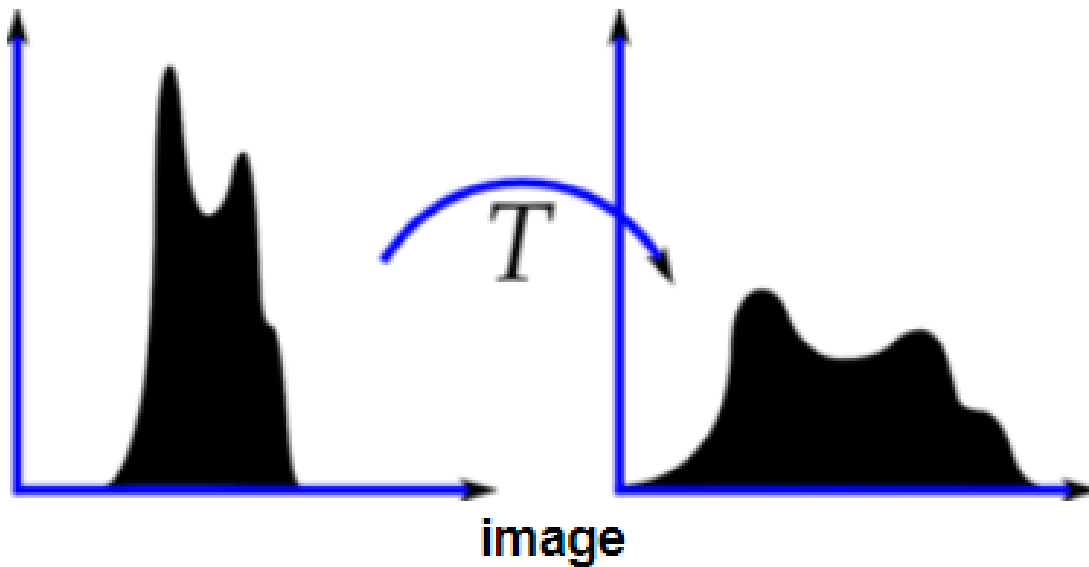
Some basic Intensity transformation Functions

Contrast stretching

Contrast stretching (also called Normalization) attempts to improve an image by **stretching the range of intensity** values it contains to make full use of possible values.

Some basic Intensity transformation Functions

Contrast stretching



Some basic Intensity transformation Functions

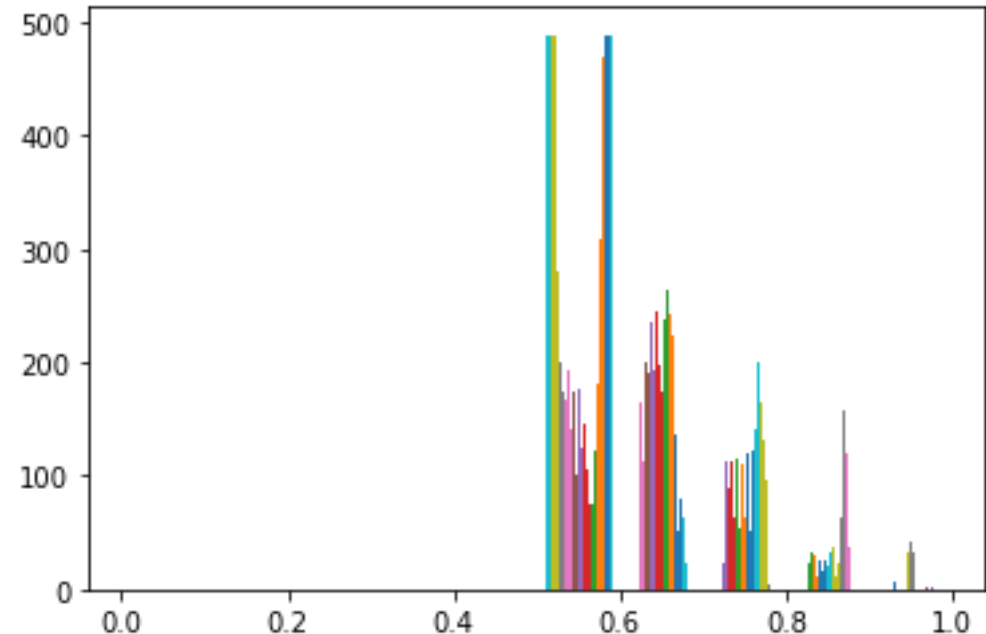
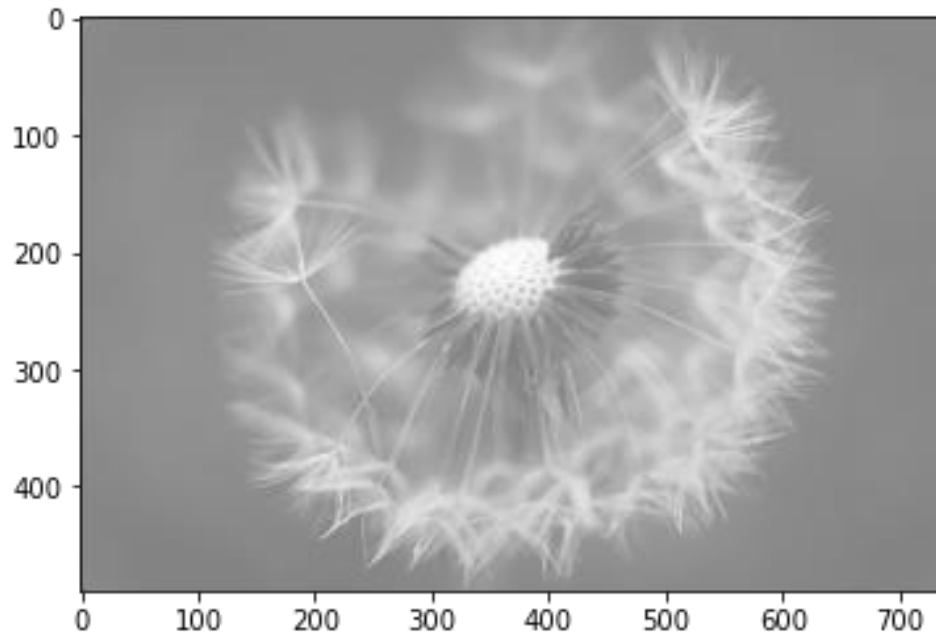
Contrast stretching

In **Min-Max Stretching**, the lower and upper values of the input image are made to span the full dynamic range.

$$X_{new} = \frac{X_{input} - X_{min}}{X_{max} - X_{min}} \times 255$$

Some basic Intensity transformation Functions

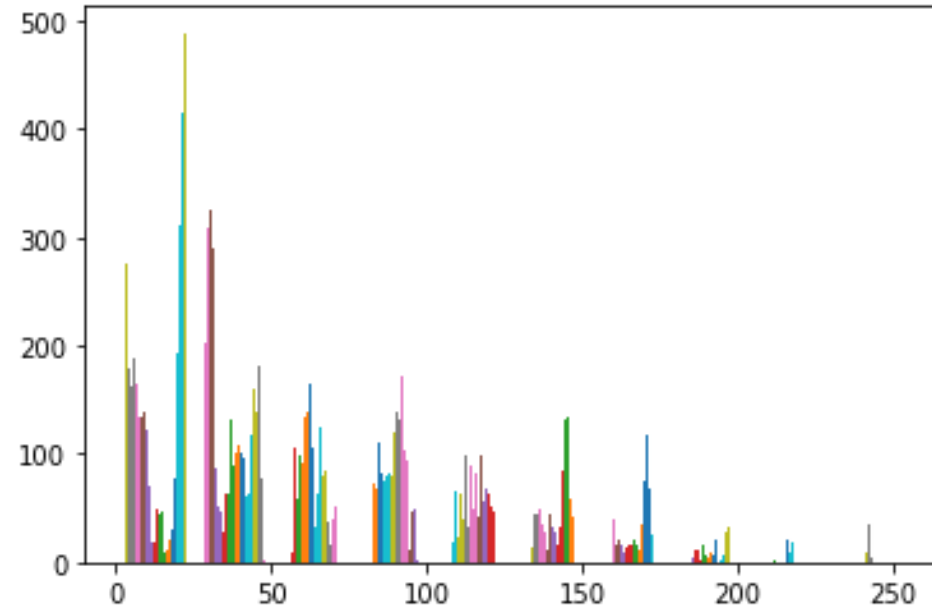
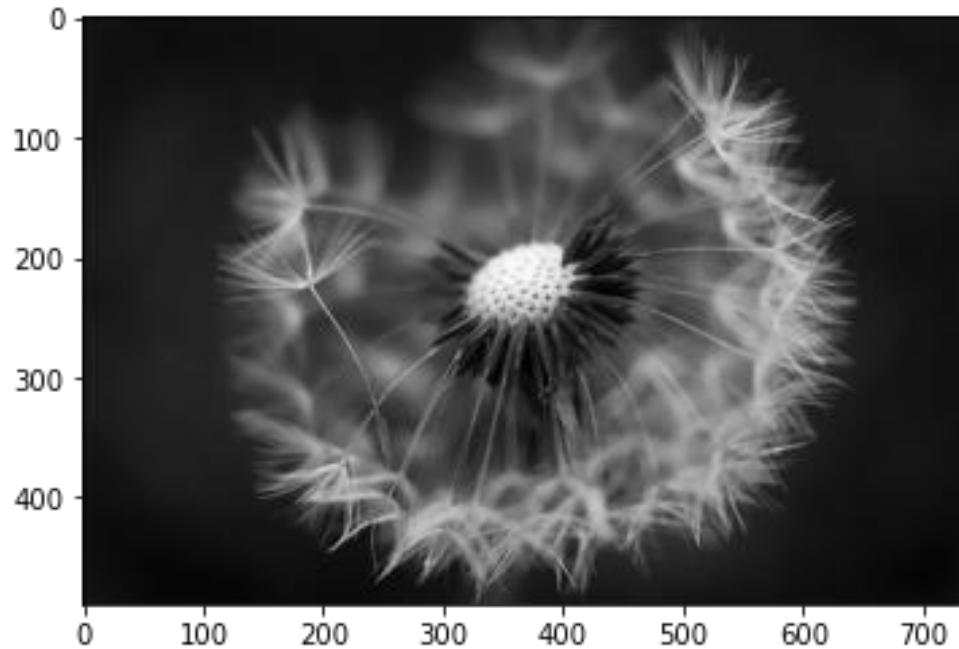
Contrast stretching



Some basic Intensity transformation Functions

Contrast stretching

Min-Max Stretching



Some basic Intensity transformation Functions

Contrast stretching

Fact to consider with `pyplot.imshow()`

```
im = plt.imread("contrast_st.png")
```

```
im = rgb_to_gray(im)  
mn = im.min()  
mx = im.max()  
print(mn, mx)
```

```
0.45877767 0.79992
```

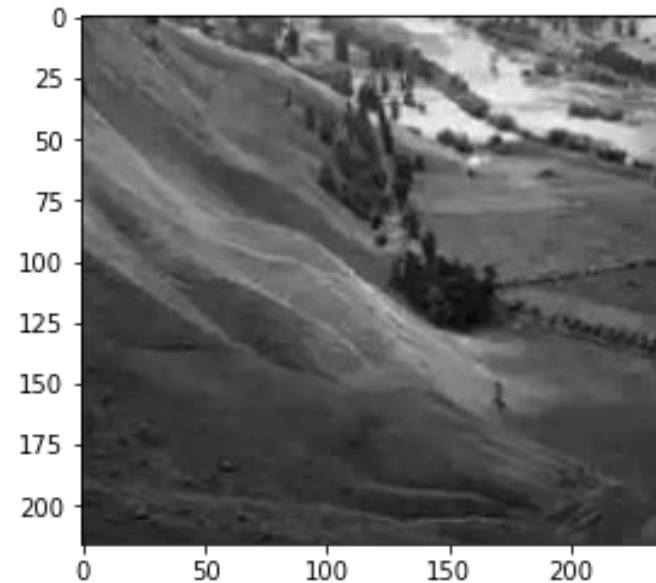
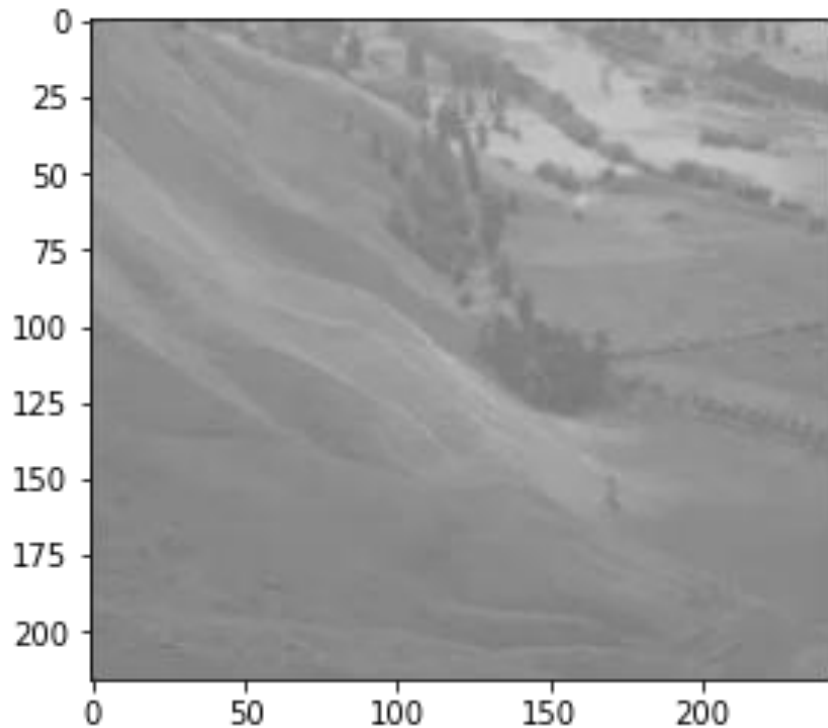


Image seems to have good contrast!!!!

Some basic Intensity transformation Functions

Contrast stretching

```
plt.imshow(im, cmap="gray", vmin=0, vmax=1)
```



Now you see the actual image

Some basic Intensity transformation Functions

Contrast stretching

One problem with this method is that outliers can reduce the effectiveness of the operation.

Suppose we have assigned some noise:

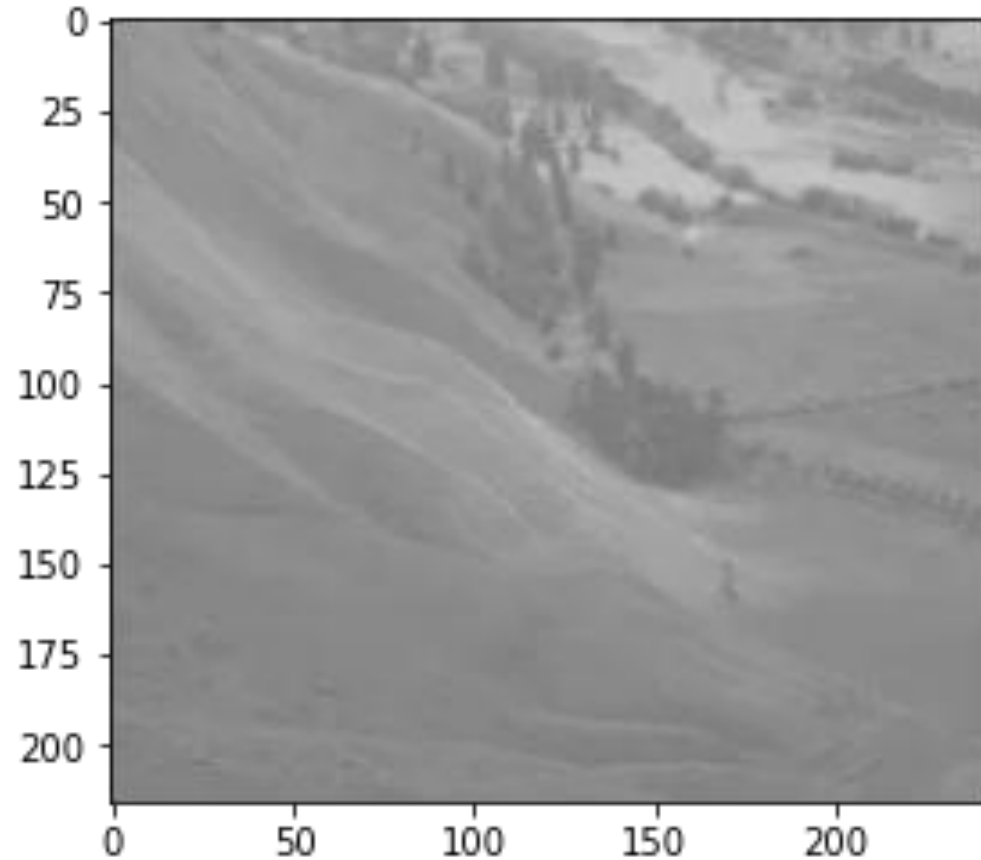
$$\text{im}[0][0] = 0.0$$

$$\text{im}[0][1] = 1.0$$

Some basic Intensity transformation Functions

Contrast stretching

No effect after
stretching



Some basic Intensity transformation Functions

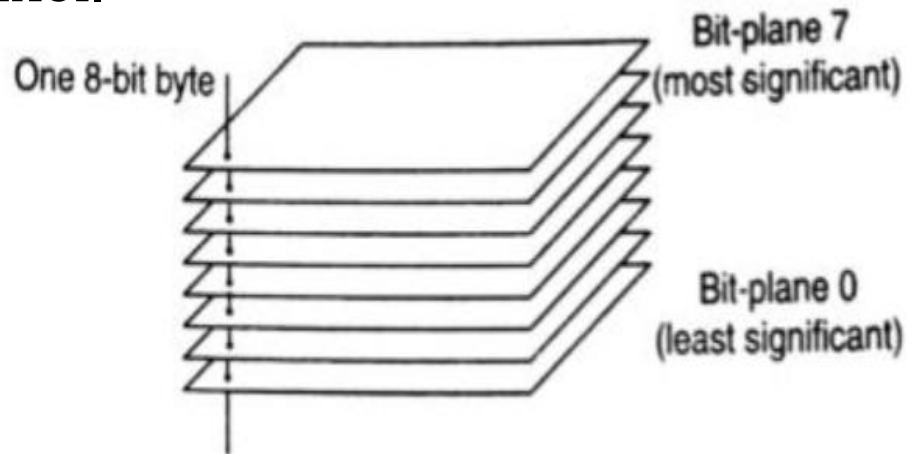
Contrast stretching

- ❑ Sometimes, when Min-Max is performed, the tail ends of the histogram becomes long resulting in no improvement in the image quality.
- ❑ So, it is better to clip a certain percentage like 1%, 2% of the data from the tail ends of the input image histogram.
- ❑ This is known as Percentile Stretching. The formulae is same as Min-Max but now the X_{max} and X_{min} are the clipped values

Some basic Intensity transformation Functions

Bit plane slicing

Bit plane slicing is a method of representing an image with one or more bits of the byte used for each pixel.



One can use only MSB to represent the pixel, which reduces the original gray level to a binary image. The three main goals of bit plane slicing is:

- ☐ Converting a gray level image to a binary image.
- ☐ Representing an image with fewer bits
- ☐ Enhancing the image by focussing.

Some basic Intensity transformation Functions

Bit plane slicing

6	7	6	6	7
0	0	0	1	2
1	1	1	2	3
4	5	5	4	2
6	6	6	7	7

110	111	110	110	111
000	000	000	001	010
001	001	001	010	011
100	101	101	100	010
110	110	110	111	111

1	1	1	1	1
0	0	0	0	0
0	0	0	0	0
1	1	1	1	0
1	1	1	1	1

MSB plane

1	1	1	1	1
0	0	0	0	1
0	0	0	1	1
0	0	0	0	1
1	1	1	1	1

Centre bit plane

0	1	0	0	1
0	0	0	1	0
1	1	1	0	1
0	1	1	0	0
0	0	0	1	1

LSB plane

Bit plane slicing

An 8-bit grayscale image



Bitplane #8



Bitplane #7



Bitplane #6



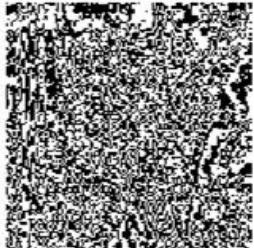
Bitplane #5



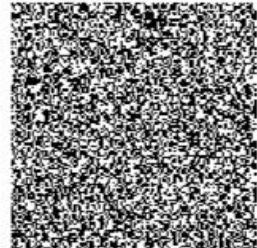
Bitplane #4



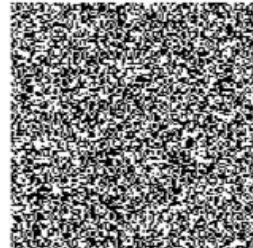
Bitplane #3



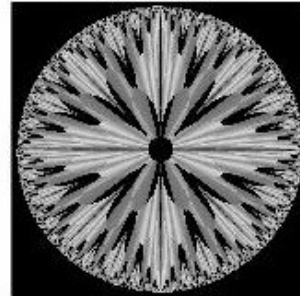
Bitplane #2



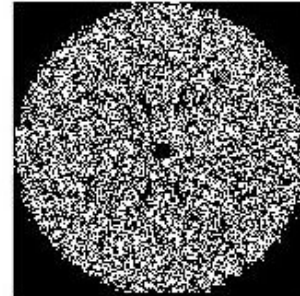
Bitplane #1



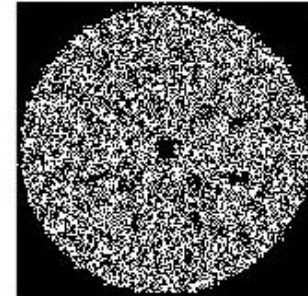
Original Image



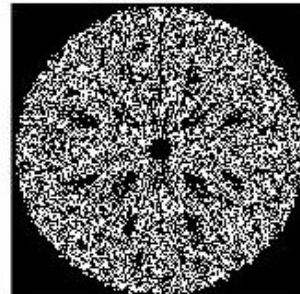
Bit plane 1



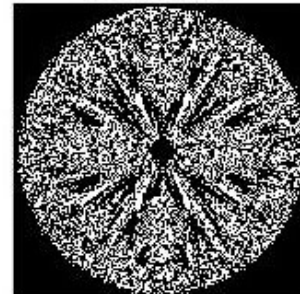
Bit plane 2



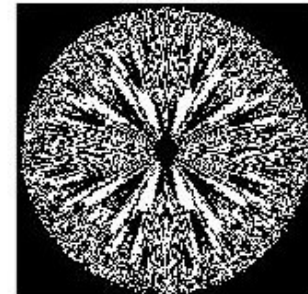
Bit plane 3



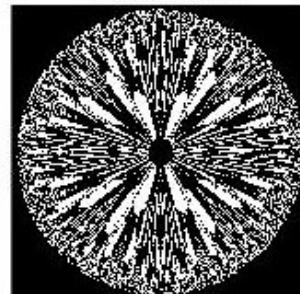
Bit plane 4



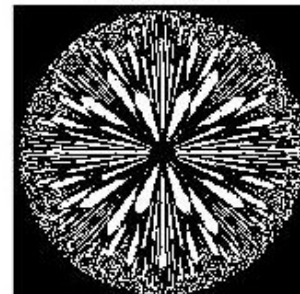
Bit plane 5



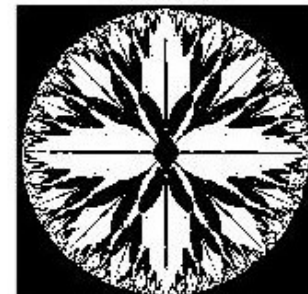
Bit plane 6



Bit plane 7



Bit plane 8



Some basic Intensity transformation Functions

Bit Plane Slicing



Histogram equalization

Histogram

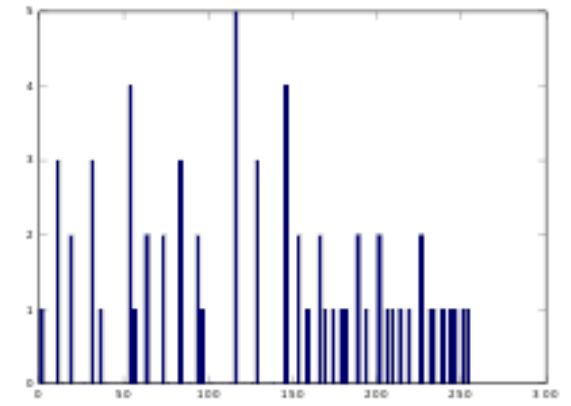
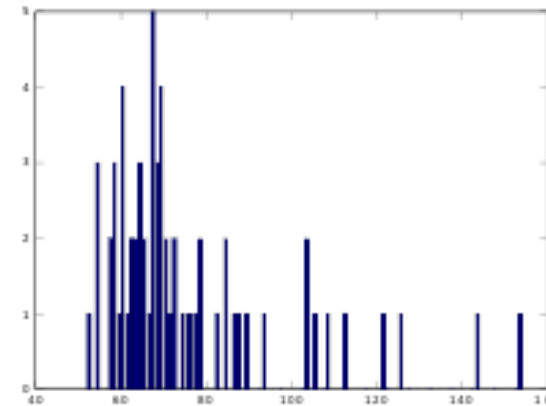
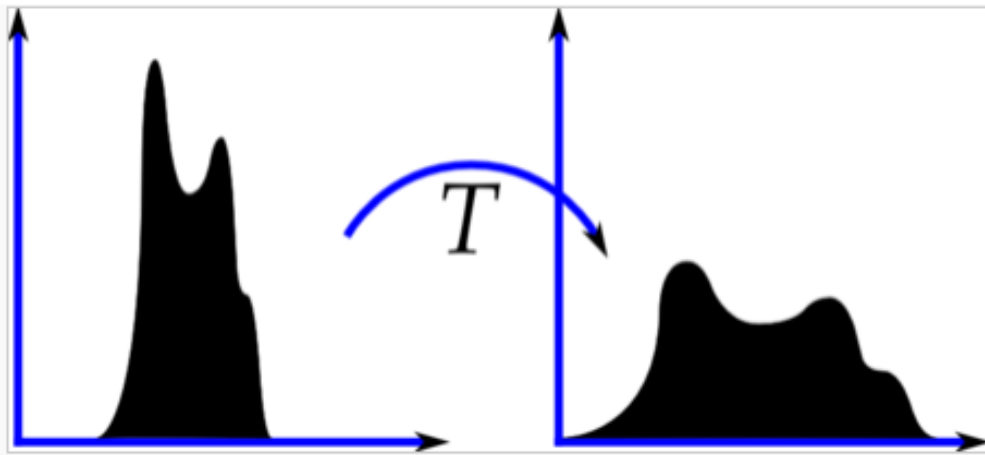
Histogram is a graphical representation of the **intensity distribution** of an image. In simple terms, it represents the **number of pixels** for **each intensity** value considered.

Histogram equalization

Histogram Equalization: It is a computer image processing technique used to improve **contrast in images**. It accomplishes this by effectively **spreading out the most frequent intensity** values, i.e. **stretching out the intensity** range of the image. This method usually increases the **global contrast** of images when its usable data is represented by **close contrast values**. This allows for areas of lower **local contrast** to gain a higher contrast.

Histogram equalization

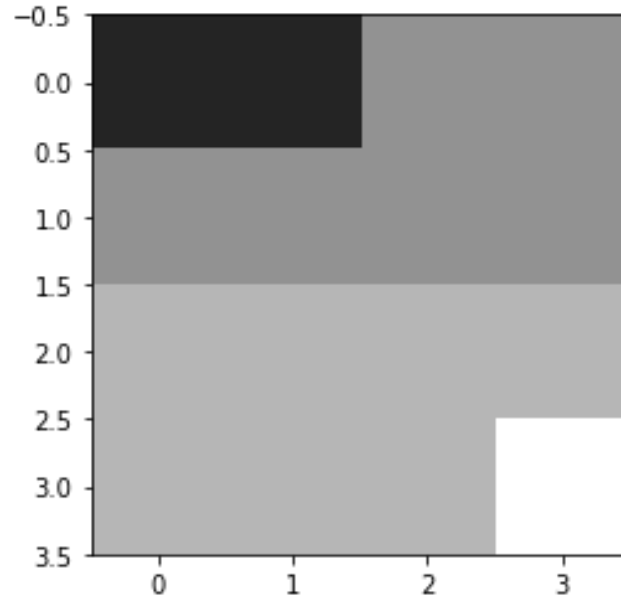
The Histogram Equalization algorithm enhances the contrast of images by transforming the values in an intensity image so that the histogram of the output image is approximately flat.



Histogram equalization

Count the frequency

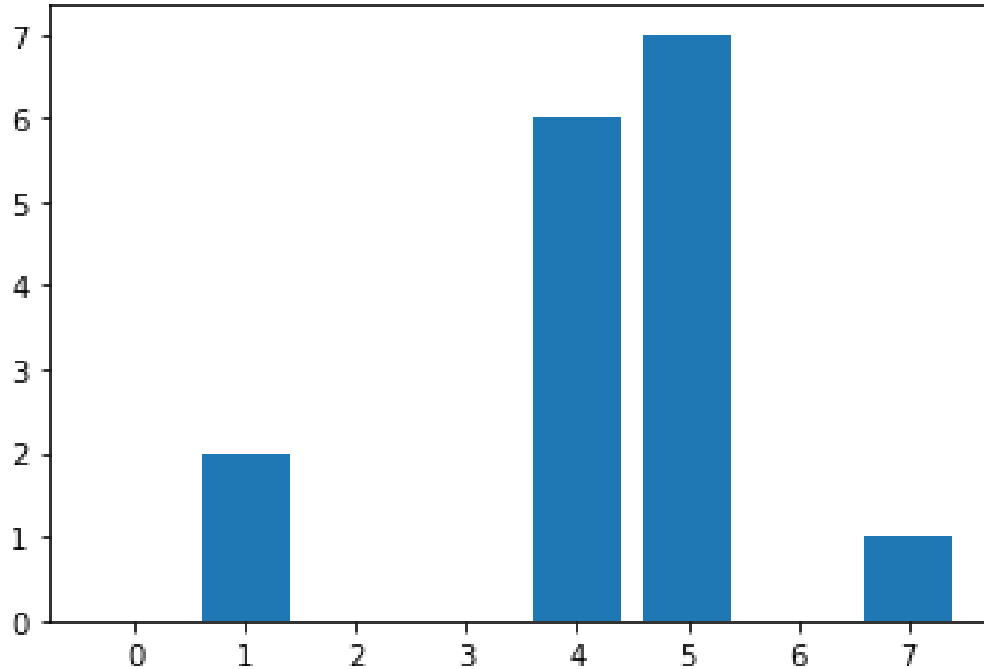
1	1	4	4
4	4	4	4
5	5	5	5
5	5	5	7



Intensity	Freq
0	0
1	2
2	0
3	0
4	6
5	7
6	0
7	1

Histogram equalization

Draw the histogram



Intensity	Freq
0	0
1	2
2	0
3	0
4	6
5	7
6	0
7	1

Histogram equalization

The Probability Density Function(PDF) defines the probability function representing the density of a continuous random variable lying between a specific range of values.

PMF - Probability mass function refers to discrete probabilities.

Histogram equalization

Calculate PMF for our intensities

Intensity	Freq	PMF (Freq/Total Pixels)
0	0	0
1	2	0.125
2	0	0
3	0	0
4	6	0.375
5	7	0.4375
6	0	0
7	1	0.0625

Histogram equalization

The cumulative distribution function (**CDF**) calculates the cumulative probability for a given x-value. It can be used as mapping function for equalization.

It is statistically proven that if we use CDF as Transformation function the PDF of the output image becomes uniform.

Histogram equalization

Calculating CDF

Intensity	Freq	PMF (Freq/Total Pixels)	CDF
0	0	0	0
1	2	0.125	0.125
2	0	0	0.125
3	0	0	0.125
4	6	0.375	0.5
5	7	0.4375	0.9375
6	0	0	0.9375
7	1	0.0625	1

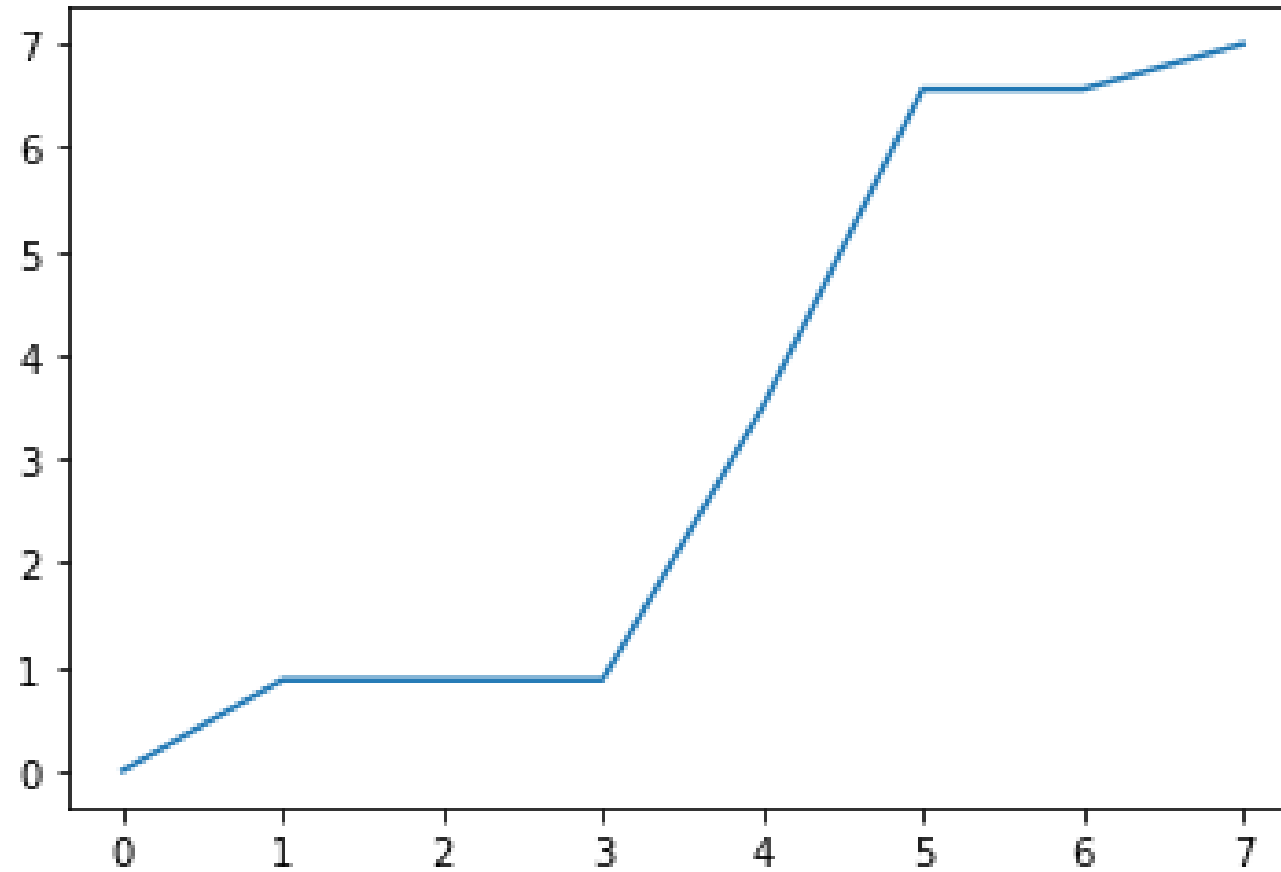
Histogram equalization

Converting CDF to required gray range

Intensity	Freq	PMF (Freq/Total Pixels)	CDF	CDF*(L-1)
0	0	0	0	0
1	2	0.125	0.125	0.875
2	0	0	0.125	0.875
3	0	0	0.125	0.875
4	6	0.375	0.5	3.5
5	7	0.4375	0.9375	6.5625
6	0	0	0.9375	6.5625
7	1	0.0625	1	7

Histogram equalization

CDF Function



Histogram equalization

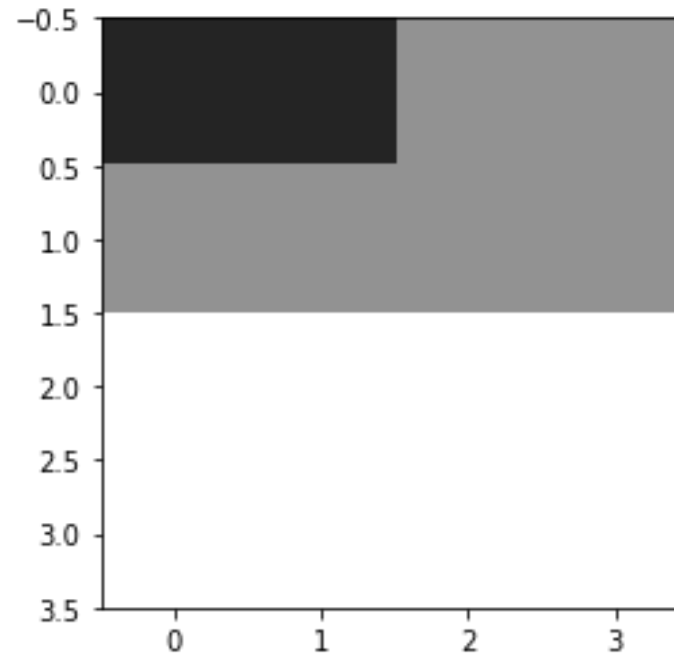
Transforming levels to nearest integers

Intensity	Freq	PMF (Freq/Total Pixels)	CDF	CDF*(L-1)	Output Intensity level
0	0	0	0	0	0
1	2	0.125	0.125	0.875	1
2	0	0	0.125	0.875	1
3	0	0	0.125	0.875	1
4	6	0.375	0.5	3.5	4
5	7	0.4375	0.9375	6.5625	7
6	0	0	0.9375	6.5625	7
7	1	0.0625	1	7	7

Histogram equalization

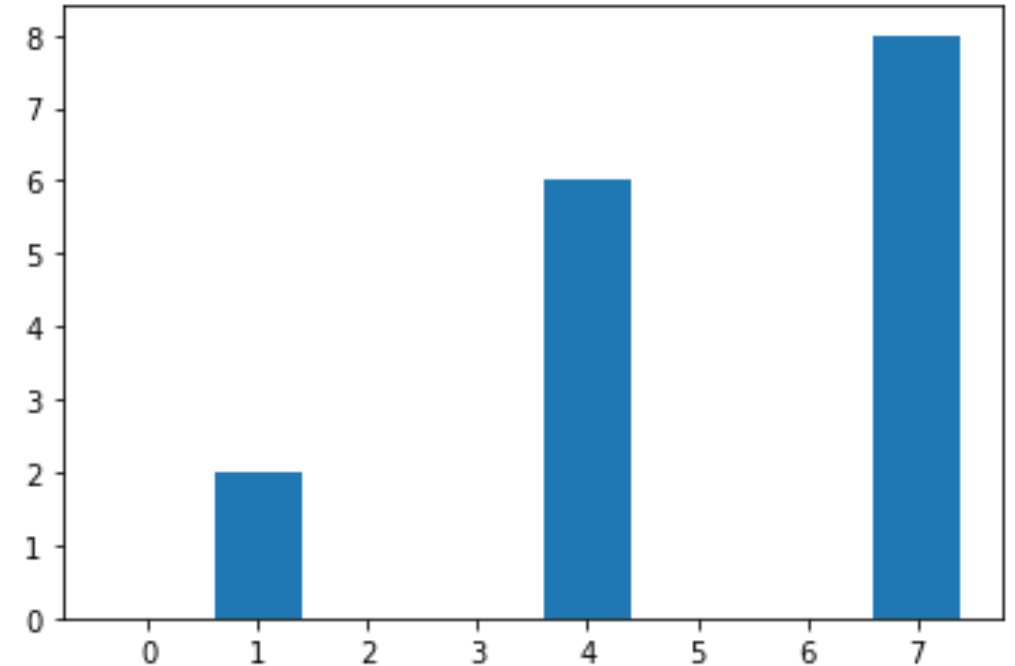
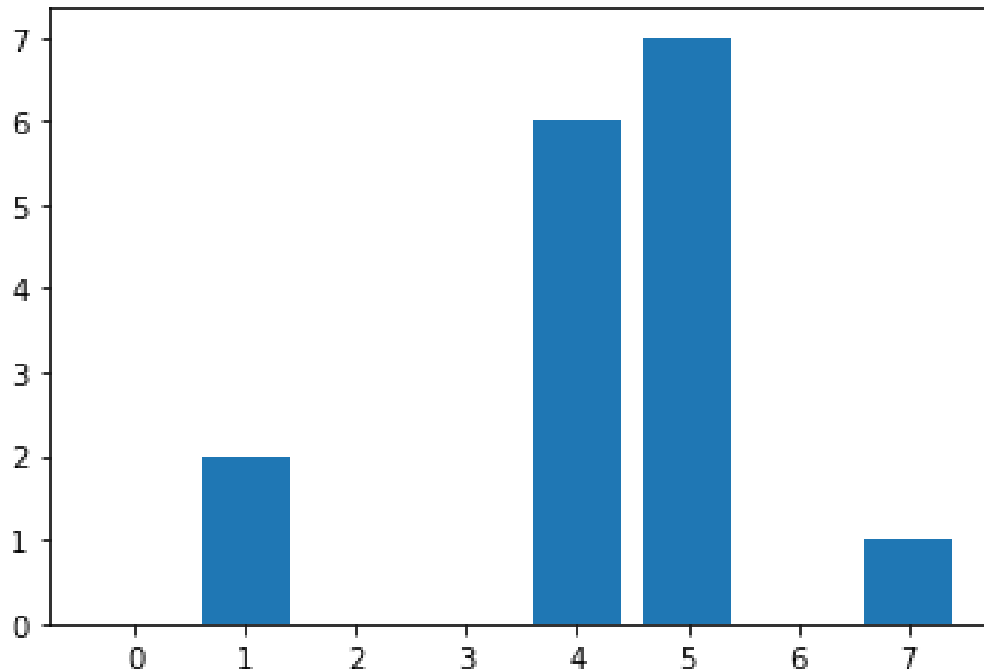
New Image

1	1	4	4
4	4	4	4
7	7	7	7
7	7	7	7



Histogram equalization

Histogram Comparison



Spatial filtering

A spatial filter consists of

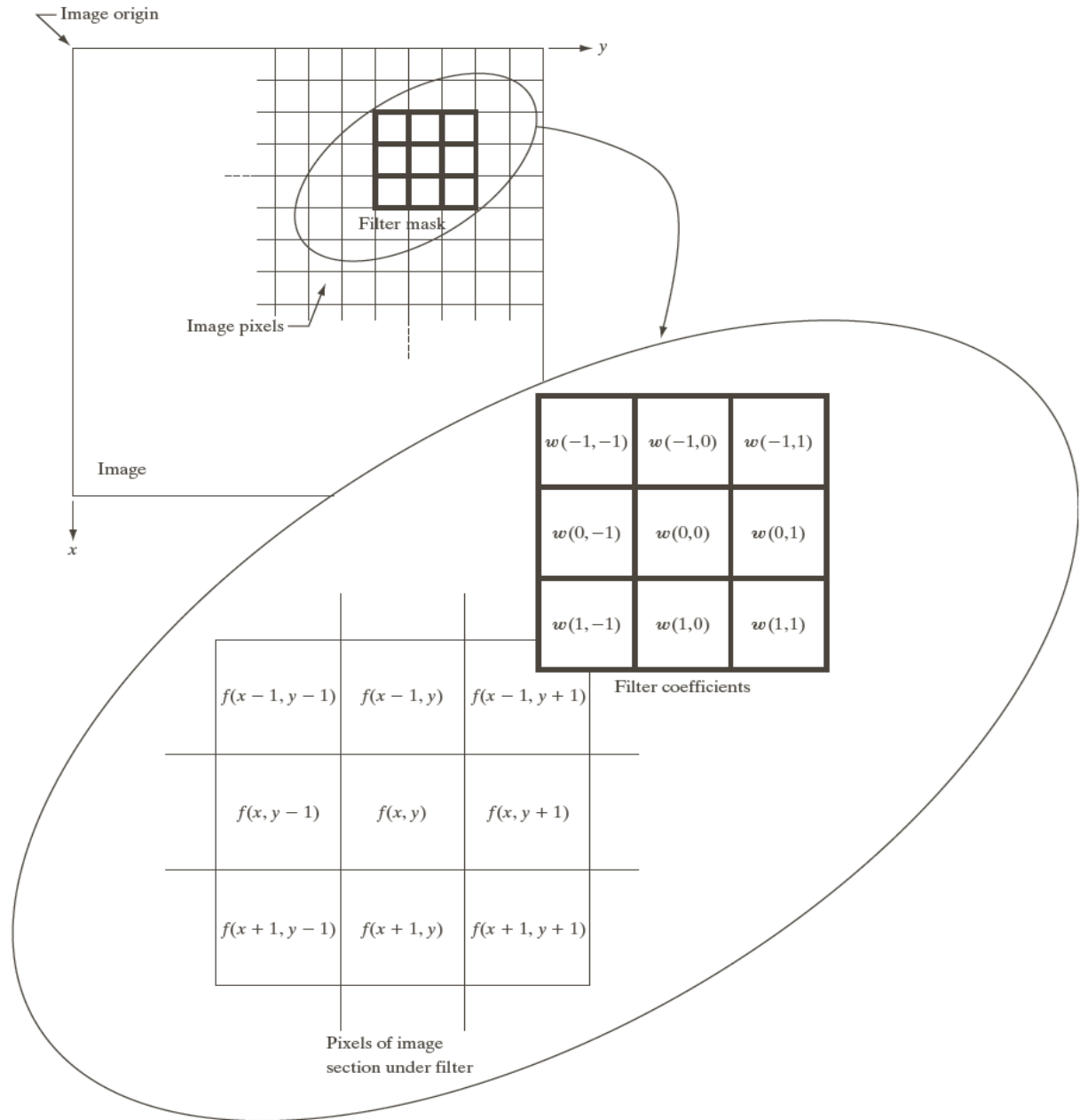
- (a) a neighborhood, and
- (b) a predefined operation

Linear spatial filtering of an image of size $M \times N$ with a filter of size $m \times n$ is given by the expression-

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

Here, $a = (m-1)/2$ and $b = (n-1)/2$

Spatial filtering



Spatial filtering

Correlation

The correlation of a filter $w(x, y)$ of size $m \times n$ with an image $f(x, y)$, denoted as $w(x, y) \star f(x, y)$

$$w(x, y) \star f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

Spatial filtering

Convolution

The convolution of a filter $w(x, y)$ of size $m \times n$ with an image $f(x, y)$, denoted as $w(x, y) \star f(x, y)$

$$w(x, y) \star f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x - s, y - t)$$

Spatial filtering

Convolution

In order to perform convolution on an image, following steps should be taken.

- Flip the mask (horizontally and vertically) only once
- Slide the mask onto the image.
- Multiply the corresponding elements and then add them
- Repeat this procedure until all values of the image has been calculated.

Spatial filtering

Convolution

25	100	75	49	130
50	80	0	70	100
5	10	20	30	0
60	50	12	24	32
37	53	55	21	90
140	17	0	23	222

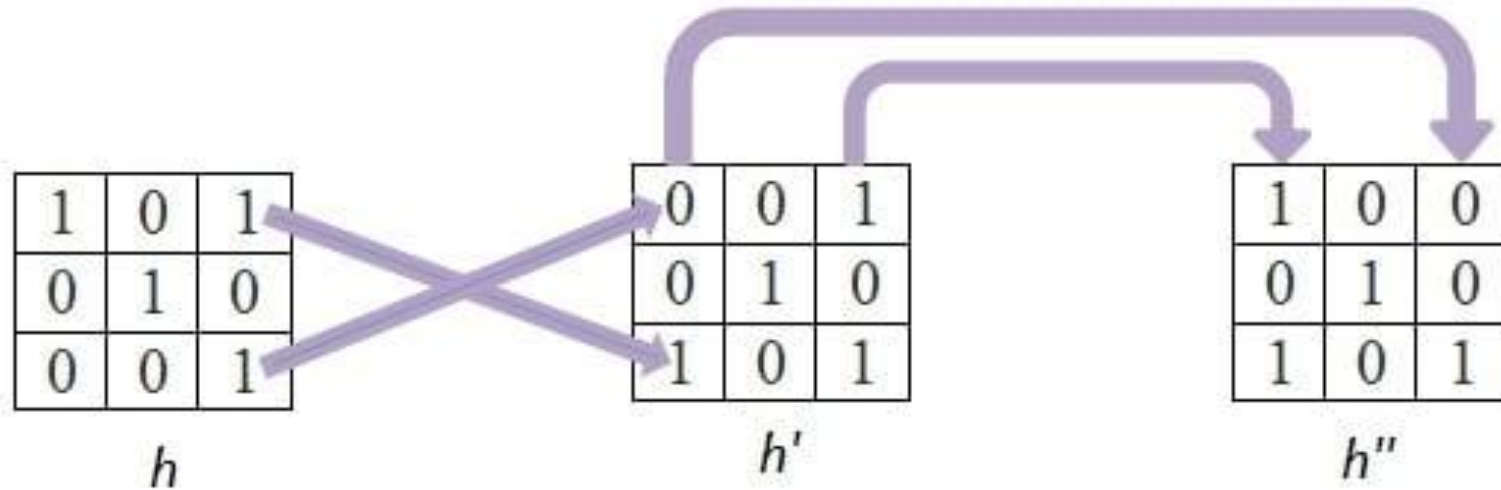
x

1	0	1
0	1	0
0	0	1

h

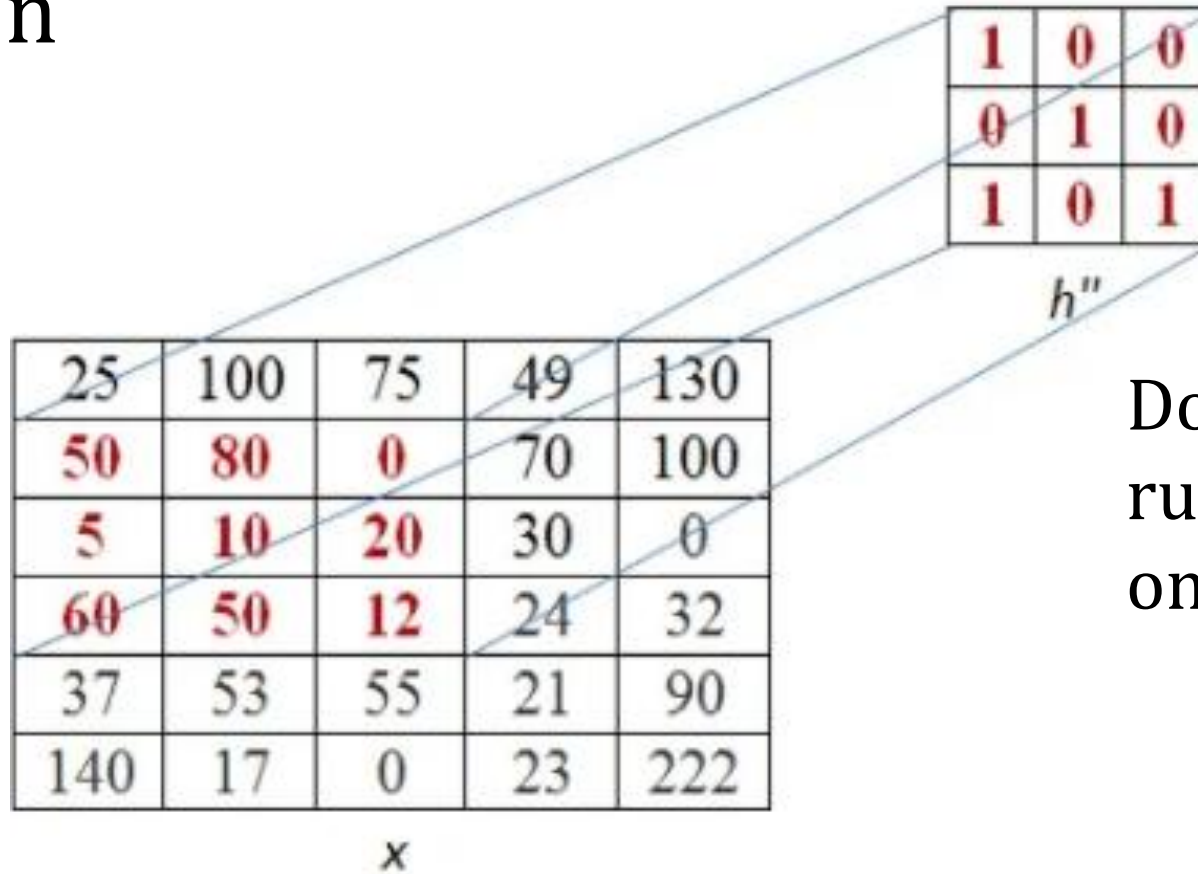
Spatial filtering

Convolution



Spatial filtering

Convolution



Do padding for
running the filter
on boundary pixels

Spatial filtering

MASK

A filter is also known as a mask. Concept of spatial filtering is also known as masking. It is done by convolving a mask with an image.

Filters are applied on image for multiple purposes. The two most common uses are as following:

- Filters are used for Blurring and noise reduction
- Filters are used or edge detection and sharpness

Spatial filtering

Smoothing Spatial Filters

- Smoothing filters are used for blurring and for noise reduction
- Blurring is used in removal of small details and bridging of small gaps in lines or curves
- Smoothing spatial filters include linear filters and nonlinear filters.

Spatial filtering

Smoothing Spatial Filters

The general implementation for filtering an $M \times N$ image with a weighted averaging filter of size $m \times n$ is given

$$g(x, y) = \frac{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)}{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t)}$$

where $m = 2a + 1$, $n = 2b + 1$.

Spatial filtering

Smoothing Spatial Filters

 $\frac{1}{9} \times$

1	1	1
1	1	1
1	1	1

Box filter

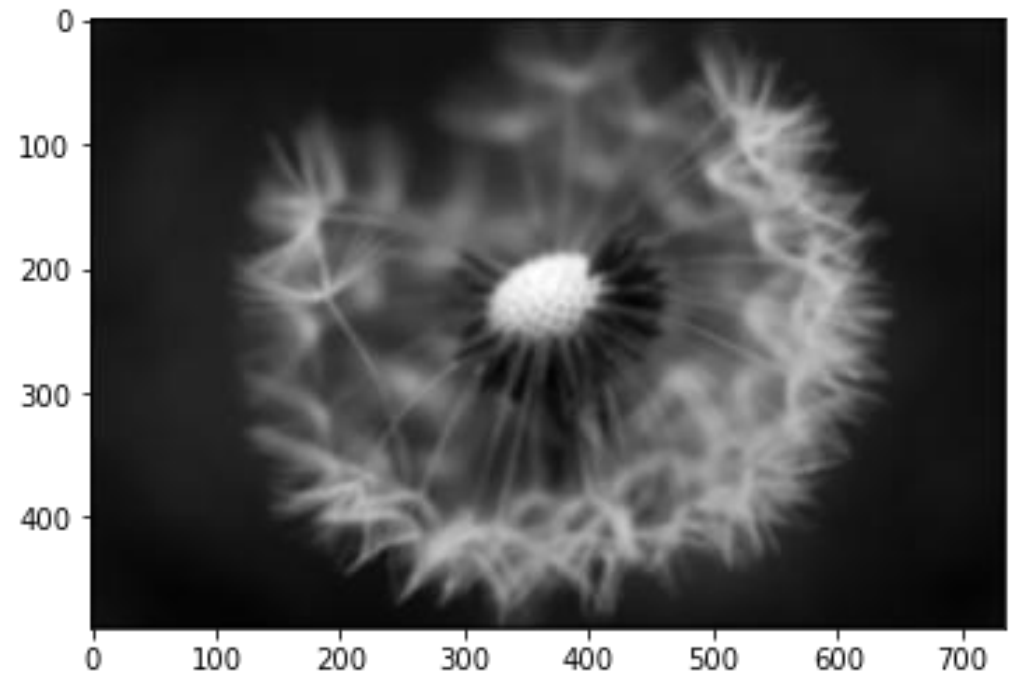
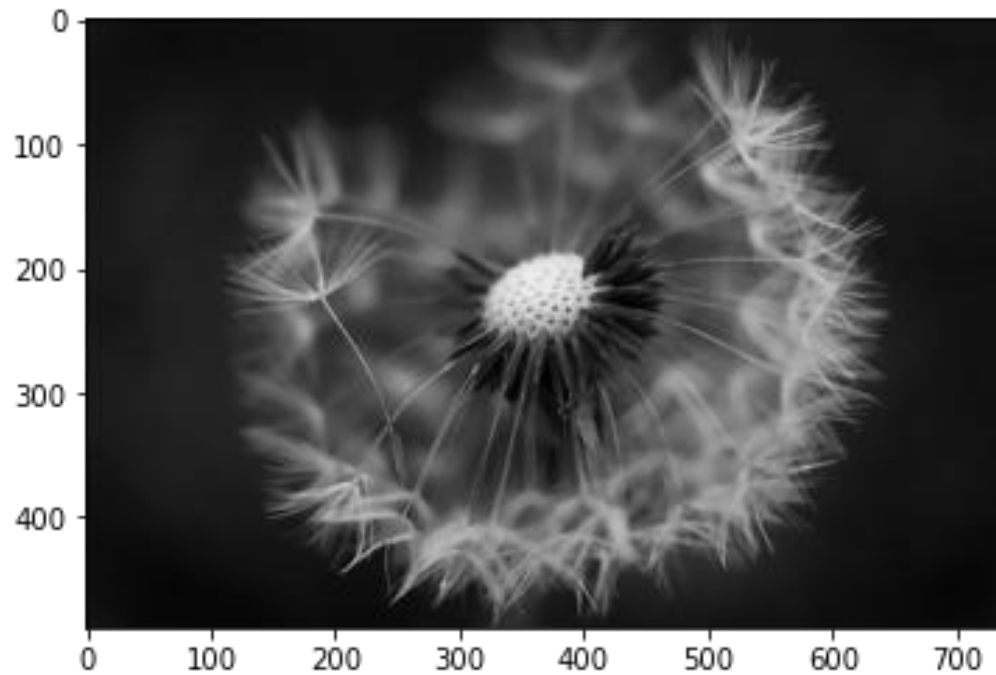
 $\frac{1}{16} \times$

1	2	1
2	4	2
1	2	1

Weighted average
filter

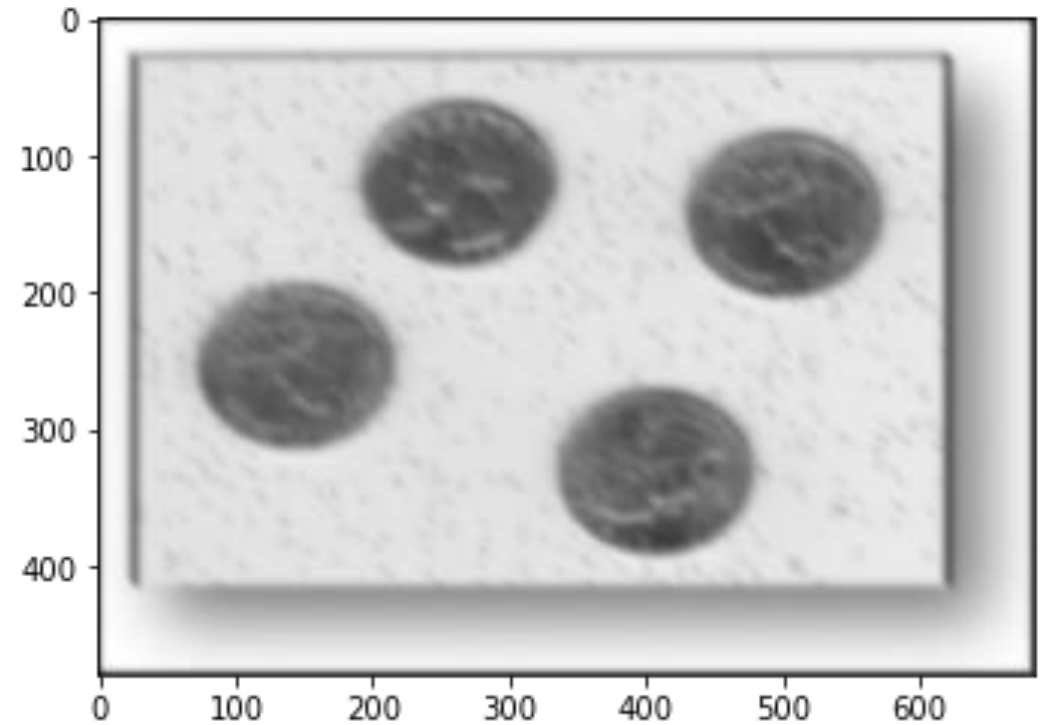
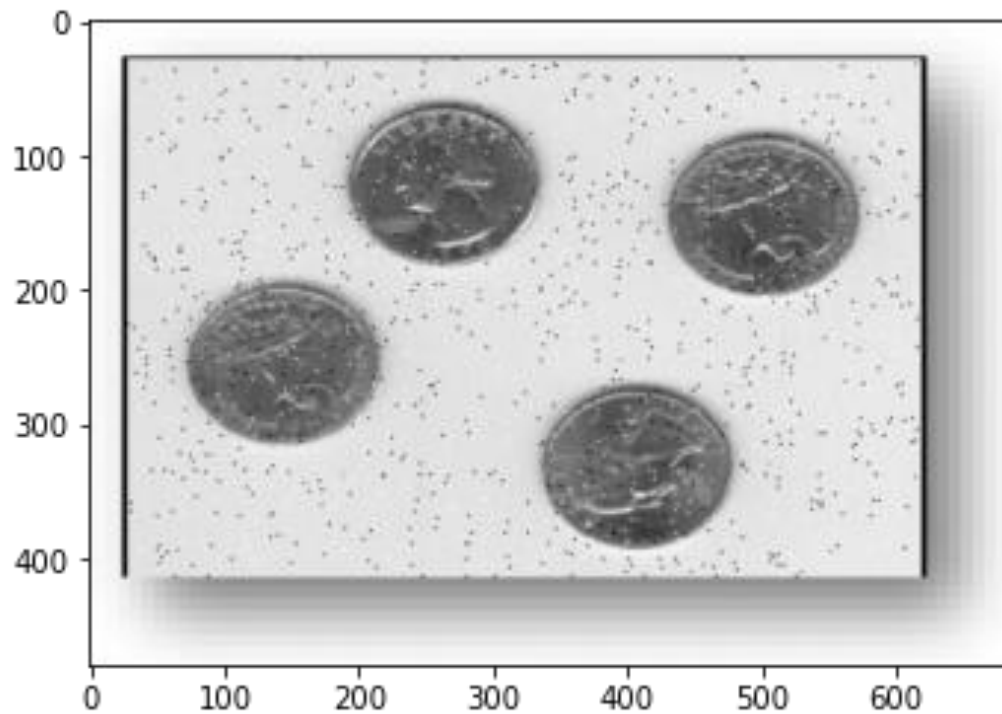
Spatial filtering

Smoothing Spatial Filters (Blurring)



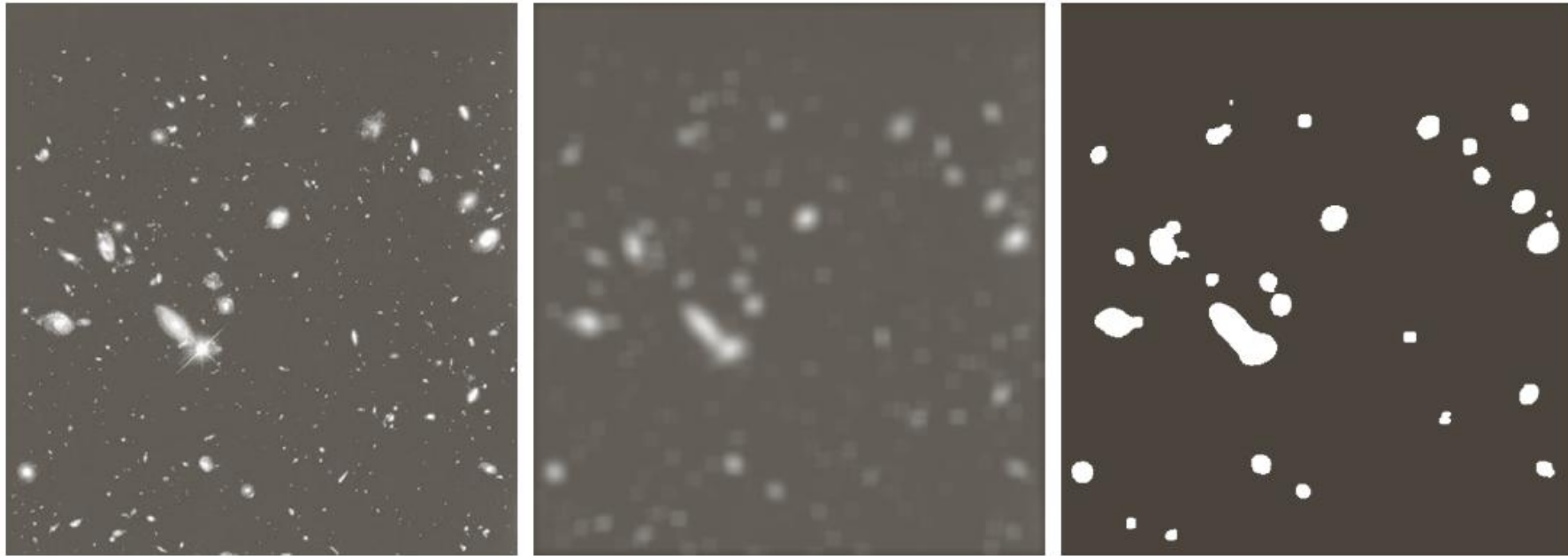
Spatial filtering

Smoothing Spatial Filters (Noise reduction)



Spatial filtering

Smoothing Spatial Filters (Noise reduction)



Spatial filtering

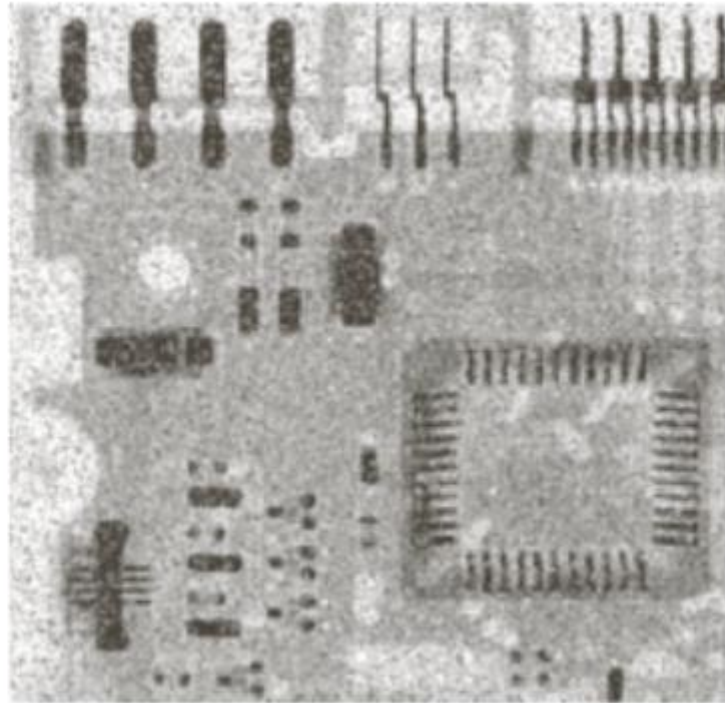
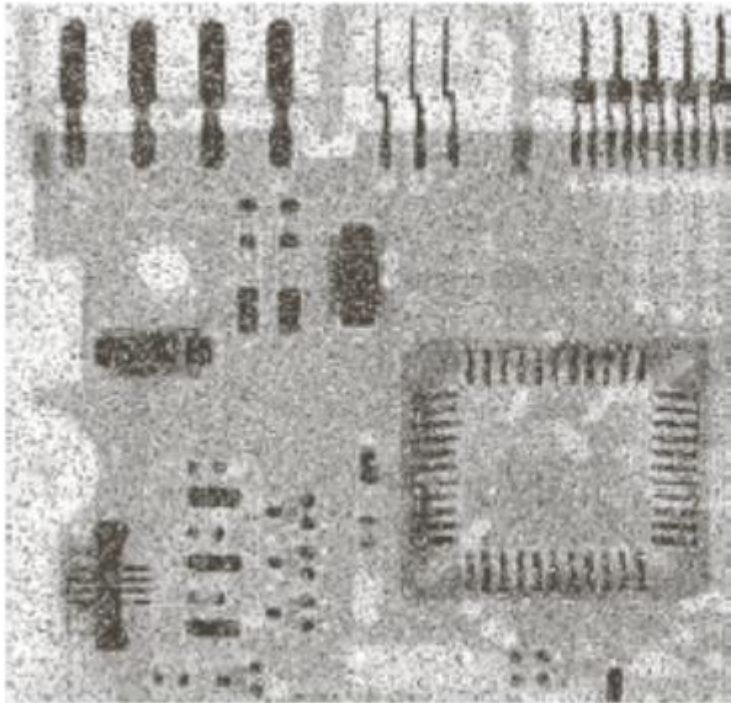
Order-statistic (Nonlinear) Filters

- Nonlinear
- Based on ordering (ranking) the pixels contained in the filter mask
- Replacing the value of the center pixel with the value determined by the ranking result

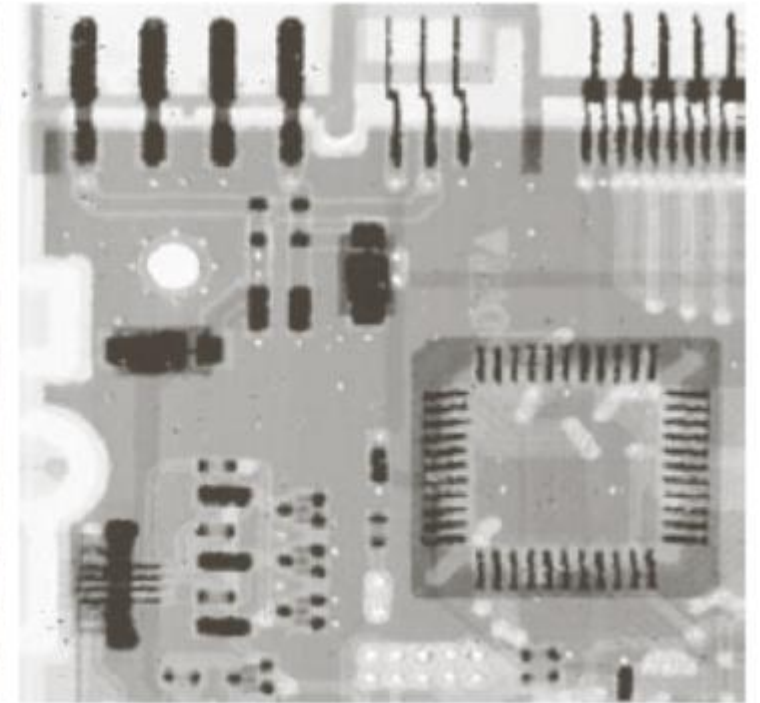
E.g., median filter, max filter, min filter

Spatial filtering

Order-statistic (Nonlinear) Filters



Mean filter



Median filter

Sharpening Spatial Filters

- ▶ Foundation
- ▶ Laplacian Operator
- ▶ Unsharp Masking and Highboost Filtering
- ▶ Using First-Order Derivatives for Nonlinear Image Sharpening
 - The Gradient

Sharpening Spatial Filters

Foundation

- ▶ The first-order derivative of a one-dimensional function $f(x)$ is the difference

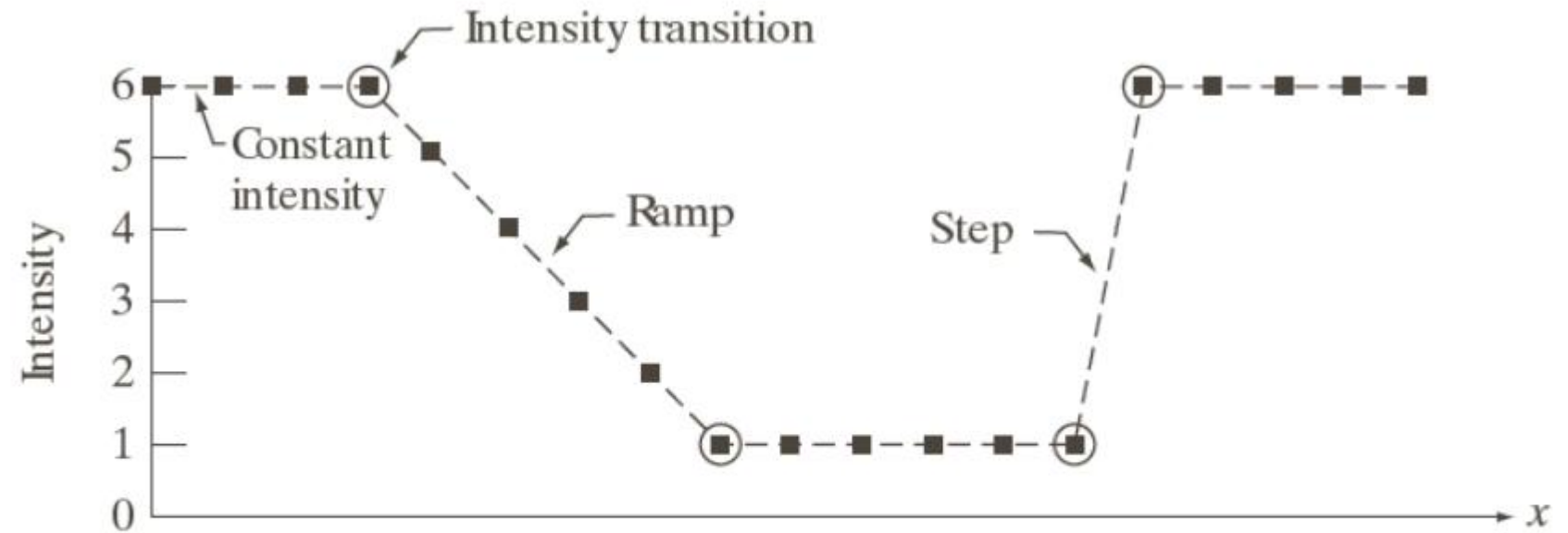
$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

- ▶ The second-order derivative of $f(x)$ as the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

Sharpening Spatial Filters

Foundation



Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6	→ x
-----------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

$$f(x+1) - f(x)$$

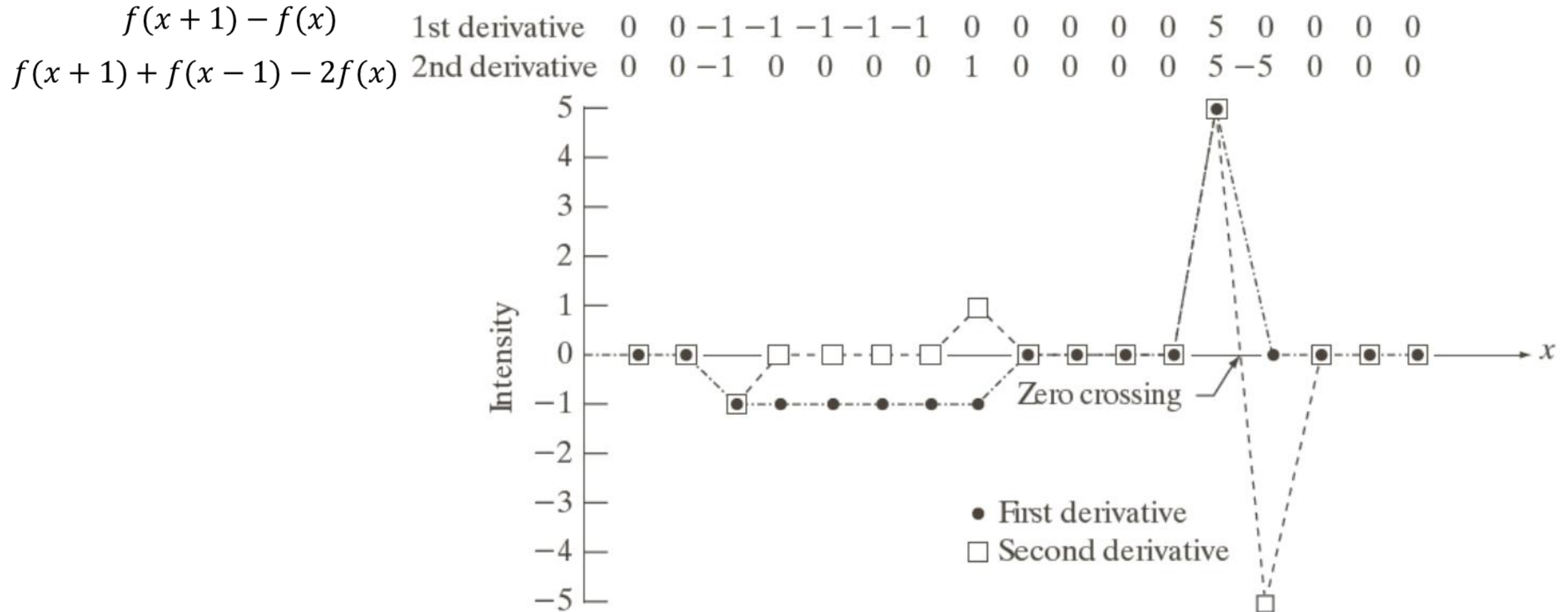
1st derivative	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0	
----------------	---	---	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---	---	--

$$f(x+1) + f(x-1) - 2f(x)$$

2nd derivative	0	0	-1	0	0	0	0	1	0	0	0	0	0	5	-5	0	0	0	
----------------	---	---	----	---	---	---	---	---	---	---	---	---	---	---	----	---	---	---	--

Sharpening Spatial Filters

Foundation



Sharpening Spatial Filters

Foundation

- ▶ First-order derivatives generally produce thicker edges in image
- ▶ Second-order derivatives have a stronger response to fine detail, such as thin lines, isolated points, and noise
- ▶ Second-order derivatives produce a double-edge response at ramp and step transition in intensity
- ▶ The sign of the second derivative can be used to determine whether a transition into an edge is from light to dark or dark to light

Sharpening Spatial Filters

The second-order isotropic derivative operator is the **Laplacian** for a function (image) $f(x,y)$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\begin{aligned} \nabla^2 f &= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) \\ &\quad - 4f(x, y) \end{aligned}$$

Sharpening Spatial Filters

Laplacian mask

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

Sharpening Spatial Filters

In Positive **Laplacian** we have standard mask in which **center** element of the mask should be negative and corner elements of mask should be zero.

In **negative** Laplacian operator we also have a standard mask, in which **center** element should be positive.

Sharpening Spatial Filters

Image sharpening in the way of using the Laplacian:

$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$

where,

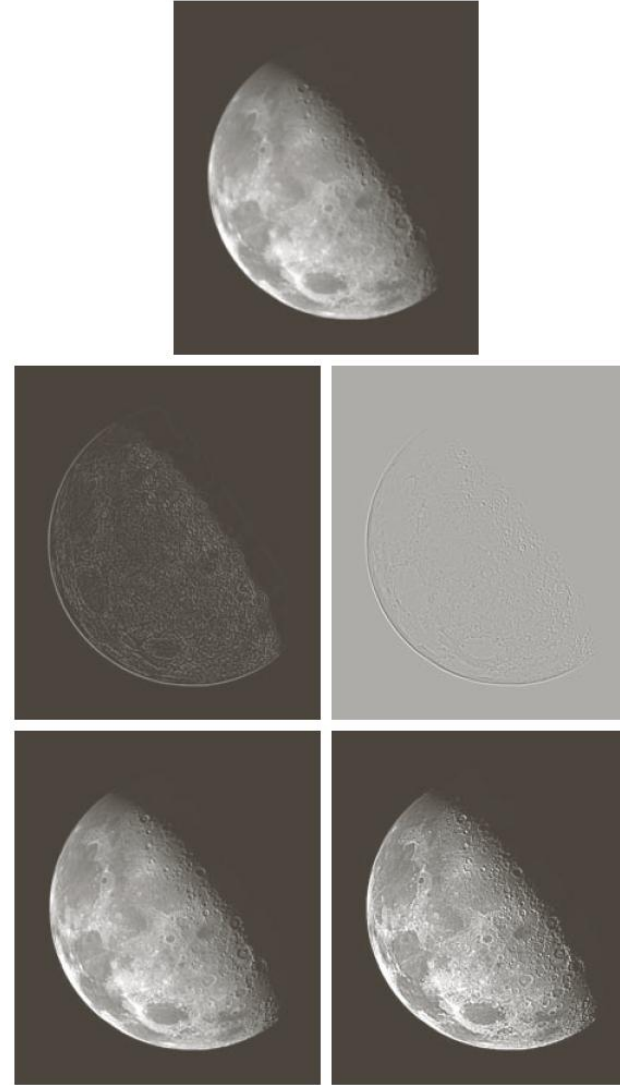
$f(x, y)$ is input image,

$g(x, y)$ is sharpenend images,

$c = -1$ if $\nabla^2 f(x, y)$ corresponding to 2 *positive mask*

Sharpening Spatial Filters

Image sharpening
in the way of
using the
Laplacian



Sharpening Spatial Filters

Unsharp Masking and Highboost Filtering

- ▶ Unsharp masking

Sharpen images consists of subtracting an unsharp (smoothed) version of an image from the original image

- ▶ Steps

1. Blur the original image
2. Subtract the blurred image from the original
3. Add the mask to the original

Sharpening Spatial Filters

Unsharp Masking and Highboost Filtering

Let $\bar{f}(x, y)$ denote the blurred image, unsharp masking is

$$g_{mask}(x, y) = f(x, y) - \bar{f}(x, y)$$

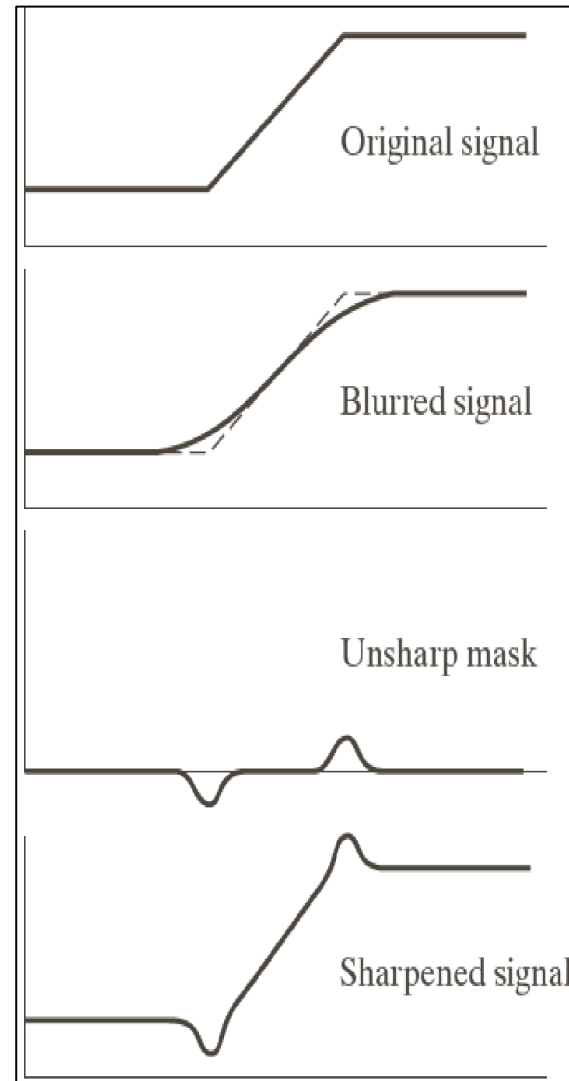
Then add a weighted portion of the mask back to the original

$$g(x, y) = f(x, y) + k * g_{mask}(x, y) \quad k \geq 0$$

when $k > 1$, the process is referred to as highboost filtering.

Sharpening Spatial Filters

Unsharp Masking and
Highboost Filtering



DIP-XE

DIP-XE

DIP-XE

DIP-XE

DIP-XE

Sharpening Spatial Filters

Other edge detection masks (study)

1. Prewitt Operator
2. Sobel Operator
3. Robinson Compass Masks
4. Krisch Compass Masks